Journal of Machine Learning Research XX (2026) 1-50 Submitted 11/26; Published –

Measuring Pairwise Diversity with Linearly Many Comparisons

Shawn Ray SHAWNRAY@CMU.EDU
Carnegie Mellon University
5000 Forbes Avenue
Pittsburgh, PA 15213, USA

Editor:

Abstract

Diversity is often measured as the average pairwise dissimilarity of n agents, models, or generated outputs. Exact evaluation needs quadratically many comparisons, and its single number hides how diversity is organized. We use the classical decomposition, or anatomy, of a pairwise table into a level, individual distinctiveness, and relational structure such as roles. The error of any subset-of-pairs average is exactly a distinctiveness term plus a relational term, and the first vanishes for every population if and only if each individual is compared equally often. With random labels, such balanced designs are unbiased, with closed-form variance set by relational structure alone. For every metric, m such comparisons have relative root-mean-square error at most $m^{-1/2}$, so $O(n + \varepsilon^{-2})$ comparisons fixed in advance give relative error ε with constant probability, an optimal rate previously reached, to our knowledge, only adaptively; uniform pair sampling can need about $n/2$ times more. On trained multi-agent teams, classifier populations, and judge-scored language-model outputs, balanced designs need up to seven times fewer comparisons than uniform pairs; with a frontier-model judge, 128 of 2,016 calls estimate the diversity of 64 responses with 1–2% root-mean-square error. The anatomy also separates two multi-agent tasks by how their diversity develops.

Keywords: diversity, incomplete U-statistics, experimental design, multi-agent reinforcement learning, evaluation of generative models

1 Introduction

Scoring every pair of 1,000 language-model responses with a learned judge takes at least 499,500 judge calls, and it produces a single number, the *average pairwise dissimilarity* of the population,

$$\mu(D) \;=\; \binom{n}{2}^{-1} \sum_{i<j} d_{ij}, \tag{1}$$

where n is the population size and the table $D = (d_{ij})$ collects the dissimilarities $d_{ij} \geq 0$ between individuals i and j. The same statistic is a standard measure of diversity across machine learning. It, or a close relative, scores the diversity of generated text and molecules (Zhu et al., 2018; Polykovskiy et al., 2020; Kirk et al., 2024) and the disagreement of ensemble members (Kuncheva and Whitaker, 2003). In multi-agent reinforcement learning (MARL), System Neural Diversity (SND; Bettini et al., 2025) is exactly (1) with d_{ij} the Wasserstein

©2026 Shawn Ray.

License: CC-BY 4.0, see https://creativecommons.org/licenses/by/4.0/. Attribution requirements are provided at http://jmlr.org/papers/vXX/XX-XXXX.html.

distance between two agents' action distributions, averaged over a set of observations, and Diversity Control (DiCo) holds it at a set point during training (Bettini et al., 2024b). Up to a factor $(n-1)/n$, the average (1) is also Rao's quadratic entropy in ecology (Rao, 1982) and nucleotide diversity in population genetics (Nei and Li, 1979). Diversity also matters for learning systems: teams of behaviorally diverse agents specialize, explore better, and are more resilient to noise and disturbances (Bettini et al., 2023, 2024b, 2025), diverse deep ensembles are better calibrated (Lakshminarayanan et al., 2017), and diverse populations of policies explore more effectively (Lehman and Stanley, 2011; Parker-Holder et al., 2020).

The average pairwise dissimilarity has two limitations that grow with the population. The first is cost. It needs $\binom{n}{2}$ comparisons, and a single comparison is often expensive: a Wasserstein distance averaged over thousands of observations, a divergence between two networks' predictions on a test set, a transport or edit distance, a rollout, or a call to a learned judge. When diversity is computed at every training iteration or fed back into a controller, this quadratic tax can become the bottleneck. The second limitation is opacity. The same average can be produced by a population in which everyone is moderately different, by one with two internally homogeneous roles, and by one with a single outlier. These are different phenomena with different consequences, and a scalar cannot tell them apart.

We show that a classical structural fact addresses both limitations. Theorem 2 (Griffing, 1956; James, 1978) states that every dissimilarity table decomposes uniquely as

$$d_{ij} \;=\; \underbrace{\mu}_{\text{level}} \;+\; \underbrace{a_i + a_j}_{\text{distinctiveness}} \;+\; \underbrace{r_{ij}}_{\text{relations}}, \qquad \textstyle\sum_i a_i = 0, \quad \sum_{j \neq i} r_{ij} = 0 \;\; \forall i, \tag{2}$$

into three mutually orthogonal parts. The level μ is the average itself. The distinctiveness a_i measures how much farther individual i is from everyone else than a typical individual is. The relational part r_{ij} is what remains once level and distinctiveness are accounted for: which particular pairs are unexpectedly similar or different, as happens with roles and clusters. We call this decomposition the *anatomy* of the table. It is canonical: the three parts are the only components that relabeling the individuals cannot mix, so whenever the compared pairs are chosen without regard to the labels, the variance of their average depends on the table only through the mean squares of the distinctiveness and relational parts (Section 3). Figure 1(a) shows the anatomy of a small team.

The anatomy turns sparse estimation of μ into a question with an exact answer. Suppose we compare only the pairs in a graph G and average them. Proposition 3 shows that the error equals a *distinctiveness leakage*, which is proportional to the covariance between how often an individual is compared and how distinctive it is, plus a *relational leakage*, which is the average residual on the compared pairs. The first term vanishes for every population if and only if every individual is compared equally often (Proposition 4). This observation leads to the designs we study, *randomized balanced designs*:[1] draw a random relabeling of the population, then compare each individual with its k neighbors in a fixed k-regular pattern (Figure 1(b)), such as a ring lattice (the *permuted circulant* of Algorithm 1) or k rounds of a round-robin schedule (the *round-robin design*). These designs have four properties.

1. Here *balanced* means that every individual is compared equally often (an equireplicate design). In the sense of balanced incomplete block designs, the only balanced design with blocks of size two is the complete graph.

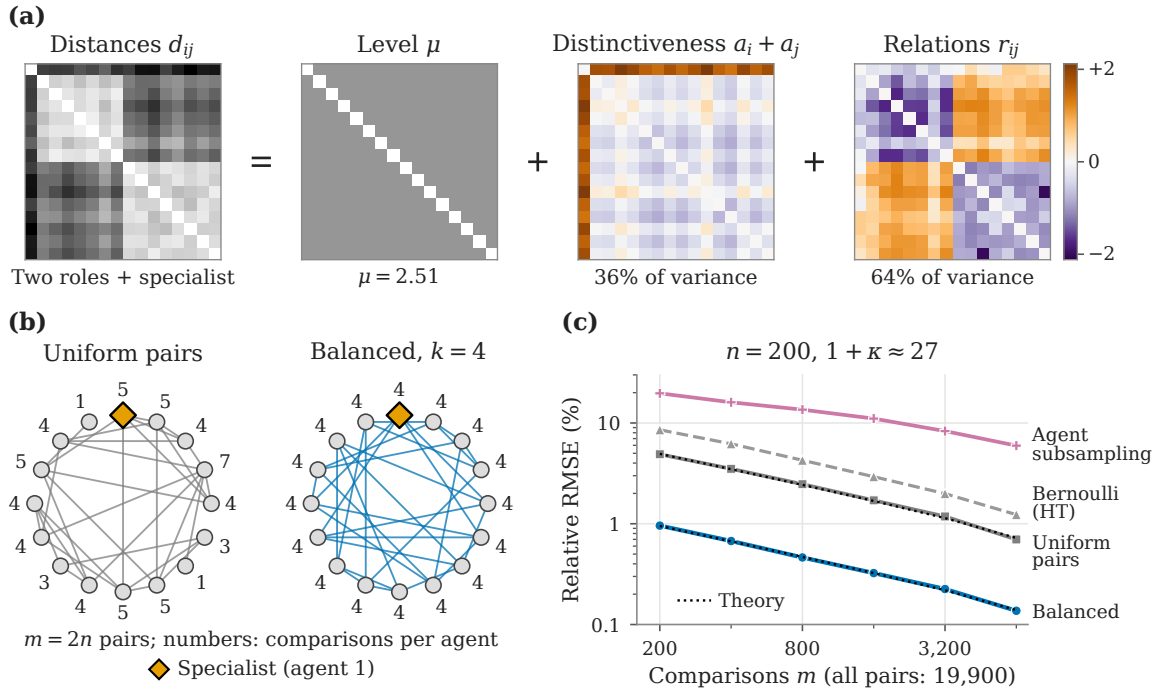


Figure 1: **The anatomy of pairwise diversity sets the cost of measuring it.** (a) A synthetic team of $n = 16$ agents with two behavioral roles and one specialist (agent 1). Its distance table splits uniquely into a level (the diversity μ), the distinctiveness table $a_i + a_j$, and relational structure r_{ij} (Theorem 2). Gray runs from 0 (white) to the largest distance; the last two panels share one symmetric color scale (orange positive, purple negative), and the percentages are their shares of the pairwise variance. (b) Two designs with the same budget of $m = 2n$ pairs. Uniform pair sampling compares one agent seven times and two agents only once; a balanced design compares every agent exactly $k = 4$ times. (c) Relative root-mean-square error (RMSE) of the estimated diversity of a synthetic 200-member population with heterogeneous distinctiveness ($1 + \kappa \approx 27$, where κ is the ratio of distinctiveness to relational energy), over 2,000 design draws, for agent subsampling (all pairs within a random subset), Bernoulli sampling with the Horvitz–Thompson (HT) estimator, uniform pairs, and the balanced design (permuted circulant). Balanced designs cancel the distinctiveness term exactly (Proposition 3), so uniform pairs have about 27 times more variance at every budget (Corollary 6); dotted lines are the closed-form predictions of Theorem 5 and Corollary 6.

- **Exact and closed-form.** The estimate is exactly unbiased, and with $m = nk/2$ comparisons its variance is $\sigma^2_{\text{int}}\,(n-1-k)/\big(m(n-3)\big) \approx 2\sigma^2_{\text{int}}/(nk)$, which depends on the population only through the relational energy σ^2_{int}, the mean squared relational residual (Theorem 5).

- **Efficiency set by the anatomy.** At equal cost, uniform pair sampling has about $1 + \kappa$ times the variance, where κ is the ratio of distinctiveness energy to relational energy; uniform sampling can win only when $\kappa = O(1/n)$, and then by a factor $1 + O(1/n)$ (Corollary 6).
- **Universal and optimal.** The triangle inequality bounds the relational energy by $(n-2)\mu^2/n$, a bound attained for every even n. Hence, for every metric, a balanced design with m comparisons has relative root-mean-square error at most $m^{-1/2}$, and $O(n + \varepsilon^{-2})$ comparisons give relative error ε with constant probability and with no further assumption on the population (Theorem 7). This rate is optimal for constant failure probability, and uniform pair sampling can require about $n/2$ times as many comparisons (Proposition 8).
- **Informative.** The same comparisons give unbiased estimates of every individual's distinctiveness (Proposition 10) and, for the round-robin design, an exactly unbiased variance estimate (Proposition 15).

For a fixed comparison graph, Theorem 9 bounds the error of any k-regular graph by its spectrum and the relational part alone, and Proposition 3 shows exactly how local interaction graphs, such as nearest-neighbor or communication graphs, under-report diversity: through negative residuals on the compared pairs and by over-representing typical individuals. Finally, Proposition 11 shows that, at each control step, a DiCo-type controller inherits the relative error of the estimate to first order.

We test these results on synthetic populations that isolate each anatomical component, on multi-agent teams trained with independent proximal policy optimization (PPO; Schulman et al., 2017; Schroeder de Witt et al., 2020) in the Vectorized Multi-Agent Simulator (VMAS; Bettini et al., 2024a), on populations of 128 and 1,024 trained classifiers, on language-model outputs scored by neural judges, and in closed-loop diversity control (Section 5). Simulations confirm the closed forms to within Monte Carlo error (Figure 1(c), Figure 2, and Section 5.6), and the anatomy predicts the gains on real populations: at equal cost, the balanced design has up to 7.2 times less variance than uniform pair sampling on trained teams (median 2.4; slightly less, within the factor of Corollary 6, when diversity is almost purely relational), 1.5–1.6 times less on classifier populations, 1.6–3.8 times less on single-prompt populations of generated text, and up to two orders of magnitude less when a few individuals are outliers; the adaptive optimal-rate scheme of Chechik et al. (2015) is 1.4–10 times less accurate than the balanced design at equal cost (Table 2). With Claude Opus 5.5 as the judge, 128 calls (about \$0.25) estimate the diversity of 64 generated responses with a relative root-mean-square error of 1.05%, whereas all 2,016 pairs cost \$3.93. One design per temperature, with its own error bar, traced a Claude-judged diversity curve over six temperatures with 1,000 responses each for \$28.80 instead of about \$6,000 (Figure 5(c)). Inside DiCo, 50 agents trained with a balanced design that uses a tenth of the comparisons per call show no detectable loss of tracking or reward (Section 5.8).

The anatomy is informative in its own right: SND reports only that diversity grows in two VMAS tasks, but the anatomy shows distinctiveness dominating in Navigation and giving way to relational structure in Dispersion, with independent or centralized critics (Section 5.4); in a classifier population the relational part recovers the four training recipes

exactly, and the most distinctive generated responses include refusals, a switch of language, and an off-topic story (Sections 5.5 and 5.6).

Contributions. Our contributions are (i) the sharp bound $\sigma^2_{\text{int}} \le (n-2)\mu^2/n$, which is a classical consequence of the triangle inequality written in anatomy coordinates and shows that a non-adaptive, equally weighted design with m comparisons has relative root-mean-square error at most $m^{-1/2}$ for every pseudometric and reaches the optimal rate (Theorem 7 and Proposition 8); (ii) for a fixed population, the exact error of every design and the anatomy-based variance of every relabeled design, which give the efficiency of balanced designs over uniform pairs and the threshold below which irregular designs do better (Proposition 3, Theorems 5 and 13, and Corollary 6); (iii) the anatomy as a diagnostic of how diversity is organized, a spectral guarantee for fixed designs that holds for every table, and a per-step tracking bound for diversity control (Section 3, Theorem 9, and Proposition 11); and (iv) experiments on synthetic populations, multi-agent teams, model populations, and language-model outputs scored by neural judges, which verify the closed forms, measure the gains over uniform and adaptive sampling on real populations, and show what the anatomy reveals (Section 5), and a NumPy implementation (Section 7).

What is classical. Several ingredients are classical. The decomposition (2) is the two-way analysis of variance of a symmetric table (Griffing, 1956) and the finite-population Hoeffding decomposition of a degree-two U-statistic (Nandi and Sen, 1963; Bloznelis and Götze, 2001), and its relational part is the U-centered matrix of distance covariance (Székely and Rizzo, 2014). The inequality behind our metric bound appears, for Euclidean data, in the proof that the distance standard deviation is at most Gini's mean difference (Edelmann et al., 2020). The variance of a uniformly relabeled design is a permutation moment of a quadratic-assignment statistic (Mantel, 1967). The permuted circulant is the circulant partial diallel cross of Kempthorne and Curnow (1961); like the round-robin pattern, it is an equireplicate design for incomplete U-statistics (Lee, 1982; Miglioli and Awan, 2025), and such designs minimize the variance for i.i.d. data (Blom, 1976; Lee, 1982). For $k = 1$ the design is a uniformly random perfect matching, whose total distance Chang (2017) analyzed in metric spaces. The sparse distinctiveness estimates rest on the textbook estimator of simple random sampling (Cochran, 1977). The lower-bound constructions follow Barhum et al. (2007), who also gave deterministic expander guarantees for ℓ_1-embeddable tables, and the rate $O(n+\varepsilon^{-2})$ was first reached, adaptively and with unequal weights, by Chechik et al. (2015); see Section 6. In survey-sampling terms, the error identity (Proposition 3) is the regression decomposition with the individual-incidence vectors as auxiliary variables (Särndal et al., 1992), regular designs are balanced samples on those vectors (Deville and Tillé, 2004), Theorem 5(iii) makes the balanced-sampling variance approximation of Deville and Tillé (2005) exact for relabeled designs, and the round-robin variance estimate is the random-group estimator (Wolter, 2007). What is new is therefore not the decomposition or its moments but their consequences for a fixed population, listed above.

Scope. Our results apply to any symmetric, non-negative dissimilarity; only the universal budget uses the triangle inequality. Squared Hilbert-space distances (squared Euclidean, cosine, or squared Hellinger distance, learned perceptual image patch similarity [LPIPS; Zhang et al., 2018], or classifier disagreement) admit an exact linear-time formula (Proposition S1 in the Supplementary Material). Our designs target the many dissimilarities without one,

such as the Wasserstein distance behind SND, transport and edit distances, and rollout- or judge-based comparisons, and need only black-box access to d.

2 Setting: Pairwise Diversity and Comparison Designs

Populations and dissimilarities. A population consists of $n \geq 4$ individuals, indexed by $[n] = \{1, \ldots, n\}$. Let $\mathcal{P} = \{\{i, j\} : 1 \leq i < j \leq n\}$ be the set of unordered pairs and $N = |\mathcal{P}| = \binom{n}{2}$. A *dissimilarity table* is a symmetric matrix $D = (d_{ij}) \in \mathbb{R}^{n \times n}$ with $d_{ii} = 0$ and $d_{ij} \geq 0$. It is a *pseudometric table* if, in addition, $d_{ij} \leq d_{il} + d_{lj}$ for all i, j, l. The *pairwise diversity* of the population is

$$\mu(D) \;=\; \frac{1}{N} \sum_{i<j} d_{ij}. \tag{3}$$

We write μ for $\mu(D)$ when the table is clear, and $S_i = \sum_{j \neq i} d_{ij}$ for the i-th row sum; $\mathbf{1}$ denotes the all-ones vector and $\mathbb{R}^{\mathcal{P}}$ the space of real tables indexed by unordered pairs. Everything in this paper is conditional on the population: D is a fixed table, and all randomness comes from which pairs we choose to compare. This is the right target when the population is the object of interest, as for a team of agents, an ensemble, or a set of checkpoints. Section 7 discusses the case where the individuals are themselves a sample from a larger distribution.

Three instances. The theory applies to three kinds of population.

- *Behavioral diversity in MARL.* Agents $i \in [n]$ act through stochastic policies $\pi_i(\cdot \mid o)$ over a shared observation and action space. Following Bettini et al. (2025), the behavioral distance is

$$d_{ij} \;=\; \frac{1}{|\mathcal{B}|} \sum_{o \in \mathcal{B}} W_2\big(\pi_i(\cdot \mid o),\, \pi_j(\cdot \mid o)\big), \tag{4}$$

  where $\mathcal{B}$ is a set of observations collected from rollouts and W_2 is the 2-Wasserstein distance, which has a closed form for Gaussian policies. With this distance, (3) is System Neural Diversity, SND $= \mu(D)$. Sharing the observation and action spaces is an assumption inherited from SND: it is what makes any two policies comparable at the same observation. Evaluating all n policies on $\mathcal{B}$ costs $O(n|\mathcal{B}|)$ forward passes, paid once by any design that touches every agent; each d_{ij} then costs $O(|\mathcal{B}|)$ distance computations, which is what a sparse design saves. The distance d is a pseudometric because W_2 is a metric. With total variation in place of W_2, as for categorical policies, μ would be computable exactly in $O(n \log n)$ time per observation and action by sorting (Proposition S1); Gaussian W_2 admits no such identity.

- *Model populations.* Individuals are trained predictors, such as ensemble members, checkpoints, or the models produced by a hyperparameter sweep. Here d_{ij} averages the Jensen–Shannon distance between their predictive distributions over an evaluation set. This is also a pseudometric.

- *Any population with an expensive comparison.* Examples include embeddings compared by Euclidean distance, sequences compared by edit distance, point clouds compared by optimal transport, and generated texts compared by a learned judge that reads both texts at once.

Comparison designs. A *comparison design* is a weighted graph $G = ([n], E, w)$ with $E \subseteq \mathcal{P}$, weights $w_e \geq 0$, and total weight $W = \sum_{e \in E} w_e > 0$. It evaluates only the pairs in E and reports

$$\widehat{\mu}_G \;=\; \frac{1}{W} \sum_{e \in E} w_e \, d_e. \tag{5}$$

The cost is $|E|$ comparisons instead of N. We write $\delta_i = \sum_{j:\{i,j\} \in E} w_{ij}$, with $w_{ij} = w_{\{i,j\}}$, for the weighted degree of i and $\bar{\delta} = 2W/n$ for its average. For an unweighted design, δ_i is the number of comparisons involving i and $W = m = |E|$. Designs can be random or fixed. The complete graph recovers μ exactly. Bernoulli sampling, which keeps each pair independently with probability p, and uniform sampling of m distinct pairs are the standard random designs. Nearest-neighbor, communication, and spatial-proximity graphs are common fixed designs. In MARL, (5) with the distance (4) was introduced as Graph-SND (Ray, 2026); our results say which graphs to use and what each of them measures.

3 The Anatomy of Pairwise Diversity

The average (3) compresses N numbers into one. To see what it hides, and what a sparse estimate of it can miss, we split the table into the parts that relabeling the individuals cannot mix.

Definition 1 (Anatomy of a dissimilarity table) *The* distinctiveness *of individual i and the* relational residual *of the pair $\{i, j\}$ are*

$$a_i \;=\; \frac{S_i - (n-1)\mu}{n-2}, \qquad r_{ij} \;=\; d_{ij} - \mu - a_i - a_j \quad (i \neq j), \tag{6}$$

with $r_{ii} := 0$ and $R = (r_{ij})$. The total, distinctiveness (main-effect), and relational (interaction) energies are

$$\sigma^2_{\text{tot}} = \frac{1}{N} \sum_{i<j} (d_{ij} - \mu)^2, \qquad \sigma^2_{\text{main}} = \frac{1}{N} \sum_{i<j} (a_i + a_j)^2, \qquad \sigma^2_{\text{int}} = \frac{1}{N} \sum_{i<j} r_{ij}^2.$$

We write $\kappa = \sigma^2_{\text{main}} / \sigma^2_{\text{int}} \in [0, \infty]$ for their ratio, with $\kappa := 0$ if $\sigma^2_{\text{main}} = 0$ and $\kappa := \infty$ if $\sigma^2_{\text{main}} > 0 = \sigma^2_{\text{int}}$.

Since $S_i/(n-1)$ is the average dissimilarity between i and the others, a_i is the excess of that average over μ, rescaled by $(n-1)/(n-2)$. A positive a_i marks an individual that is far from everyone, such as a specialist or an outlier. A negative a_i marks a central, typical individual. The residual r_{ij} is positive for pairs that are more different than their distinctiveness predicts and negative for pairs that are more alike, as happens within a role. Since $r_{ij} = d_{ij} - (S_i + S_j)/(n-2) + n\mu/(n-2)$, the residual table R is exactly the

U-centered matrix of D used for unbiased distance covariance (Székely and Rizzo, 2014), and $\sigma^2_{\text{int}} = \frac{n-3}{n-1}\,\Omega_n$, where $\Omega_n = \frac{1}{n(n-3)}\sum_{i\neq j} r^2_{ij}$ is their unbiased estimator of the squared distance variance when D is a table of distances between sample points.

The decomposition itself is classical (Section 6); we state it in the form used throughout.

Theorem 2 (Anatomy of diversity; Griffing, 1956; James, 1978) *Every dissimilarity table D satisfies the following.*

(i) Existence and uniqueness. *$d_{ij} = \mu + a_i + a_j + r_{ij}$ for all $i \neq j$, with $\sum_i a_i = 0$ and $R\mathbf{1} = 0$. Conversely, if $d_{ij} = \mu' + a'_i + a'_j + r'_{ij}$ for all $i \neq j$, with $\sum_i a'_i = 0$, $r'_{ii} = 0$ for all i, and $R'\mathbf{1} = 0$, then $(\mu', a', R') = (\mu, a, R)$.*

(ii) Energy identity. *The level, distinctiveness, and relational components are mutually orthogonal in $\mathbb{R}^{\mathcal{P}}$. Hence $\sigma^2_{\text{tot}} = \sigma^2_{\text{main}} + \sigma^2_{\text{int}}$, and $\sigma^2_{\text{main}} = \frac{n-2}{N}\sum_i a_i^2$.*

(iii) Canonicity. *Under relabeling of the individuals by the symmetric group $\mathfrak{S}_n$, the space $\mathbb{R}^{\mathcal{P}}$ splits into exactly three irreducible invariant subspaces, of dimensions 1, $n-1$, and $n(n-3)/2$. They are the constants, the main-effect tables $\{(a_i + a_j)_{i<j} : \sum_i a_i = 0\}$, and the residual tables $\{R : R\mathbf{1} = 0\}$. The decomposition is multiplicity-free, so it is the only decomposition of $\mathbb{R}^{\mathcal{P}}$ into irreducible invariant subspaces.*

Parts (i) and (ii) are a two-way analysis of variance for a symmetric table with an empty diagonal. They match the Hoeffding decomposition of a U-statistic under sampling without replacement from a finite population (Nandi and Sen, 1963; Bloznelis and Götze, 2001) and the symmetric social relations model of psychology (Kenny and La Voie, 1984). Part (iii) explains why this decomposition, and no other, governs the error of every relabeling-invariant design: the space of pairwise tables is the permutation module $M^{(n-2,2)}$ of $\mathfrak{S}_n$, which splits into the Specht modules $S^{(n)} \oplus S^{(n-1,1)} \oplus S^{(n-2,2)}$ (James, 1978; Diaconis, 1988), and by Schur's lemma any linear operator that commutes with relabeling, such as the second-moment operator of a uniformly relabeled design, acts as a scalar on each part. The variance of every such design is therefore a combination of σ^2_{main} and σ^2_{int} with coefficients set by the design (Theorem 13 and Remark 14 in Appendix A.5). In the language of designed experiments, these are the strata of the triangular association scheme on pairs (Bose and Shimamoto, 1952), and the dependence of a randomized design's variance on stratum energies is the randomization theory of Bailey (1991, 2004). Proofs are in Appendix A, and Table S1 in the Supplementary Material summarizes the notation.

Diagnostics. The anatomy turns one number into a profile: the *level* μ; the *distinctiveness share* $\sigma^2_{\text{main}}/\sigma^2_{\text{tot}}$ and profile $(a_i)_{i\in[n]}$, which say whether diversity is carried by a few individuals and by which ones; the *relational share* $\sigma^2_{\text{int}}/\sigma^2_{\text{tot}}$ and the spectrum of R, which reveal roles or clusters; and the scale-free ratios σ_{main}/μ and σ_{int}/μ, which, as we show next, set the cost of measuring μ. Three populations can share the same μ while having very different anatomies: one with $\sigma^2_{\text{main}} \approx \sigma^2_{\text{int}} \approx 0$ (everyone equally different), one with $\sigma^2_{\text{main}}/\sigma^2_{\text{tot}} \approx 1$ (a few outliers), and one with $\sigma^2_{\text{int}}/\sigma^2_{\text{tot}} \approx 1$ (roles). Figure 1(a) shows a team that has all three components. With K well-separated roles, $R \approx F - \text{Diag}(F)$ (with equality when d_{ij} depends only on the roles of i and j), where the role-level table F (constant on blocks of individuals that share a role) has rank at most K, and effectively $K-1$ because the constraint $R\mathbf{1} = 0$ nearly centers it. Because of its zero diagonal, R generically

has rank $n-1$, but its $K-1$ leading eigenvectors indicate role membership, as in classical scaling (Torgerson, 1952; Gower, 1966).

4 Which Pairs Should We Compare?

We first derive the exact error of any comparison design, and then use it to construct randomized balanced designs (Section 4.2), give a budget that holds for every metric (Section 4.3), analyze fixed designs (Section 4.4), estimate each individual's distinctiveness from the same comparisons (Section 4.5), and show how estimation error propagates into closed-loop diversity control (Section 4.6).

4.1 An exact error identity

Substituting the anatomy into the design estimator (5) gives its error in closed form.

Proposition 3 (Exact error identity) *For every dissimilarity table D and design G,*

$$\widehat{\mu}_G - \mu \;=\; \underbrace{\frac{1}{W}\sum_{i=1}^{n}(\delta_i - \bar{\delta})\, a_i}_{\text{distinctiveness leakage}} \;+\; \underbrace{\frac{1}{W}\sum_{e\in E} w_e\, r_e}_{\text{relational leakage}}. \tag{7}$$

The level cancels because $\widehat{\mu}_G$ is a weighted average. The distinctiveness leakage is proportional to the covariance between how often an individual is compared and how distinctive it is. The relational leakage is the weighted average residual over the compared pairs. The identity is exact, holds for every design, random or fixed, and needs no assumption on D. It shows at once what each common design gets wrong.

- A *nearest-neighbor or communication graph* compares individuals that are close in some space. If proximity predicts behavioral similarity, the compared pairs have negative residuals, and the relational leakage is negative: the design systematically under-reports diversity. Individuals with many neighbors also dominate the average, so the design leaks distinctiveness as well. Such a design is better read as a measure of *local* diversity than as an estimate of μ (Section 4.4).
- *Uniform or Bernoulli pair sampling* is unbiased, but its degrees fluctuate. Each individual's random degree multiplies its distinctiveness, which adds variance proportional to σ^2_{main}.
- *Agent subsampling*, which compares all pairs within a random subset of s individuals, gives degree $s-1$ to the chosen individuals and 0 to the rest. Its distinctiveness leakage is large, and its variance decays only like $m^{-1/2}$ in the number of comparisons.

The distinctiveness leakage, the first term of (7), can be removed completely.

Proposition 4 (Balance is necessary and sufficient) *Fix a design G and $\alpha > 0$, and let $\mathcal{D}_\alpha$ be the set of pseudometric tables whose anatomy has $R = 0$ and $\|a\|_2 \le \alpha$. Then*

$$\sup_{D\in\mathcal{D}_\alpha} |\widehat{\mu}_G - \mu| \;=\; \frac{\alpha\,\|\delta - \bar{\delta}\mathbf{1}\|_2}{W}.$$

Algorithm 1 Balanced pairwise diversity (one design; repeat L times for intervals or high-probability guarantees)

Require: population $[n]$; dissimilarity oracle $d(\cdot,\cdot)$; degree $1 \le k \le n-1$ with nk even

Draw a uniformly random bijection $\sigma : \mathbb{Z}_n \to [n]$

$E \leftarrow \big\{\{\sigma(t), \sigma(t+s)\} : t \in \mathbb{Z}_n,\ 1 \le s \le \lfloor k/2 \rfloor\big\}$ ▷ permuted circulant

if k is odd **then** $E \leftarrow E \cup \big\{\{\sigma(t), \sigma(t+n/2)\} : 0 \le t < n/2\big\}$ ▷ n is even here

Evaluate d_e for all $m = nk/2$ pairs $e \in E$ ▷ the only expensive step; parallel

$T \leftarrow \frac{1}{m}\sum_{e\in E} d_e$ ▷ diversity estimate

$\widehat{m}_i \leftarrow \frac{1}{k}\sum_{j:\{i,j\}\in E} d_{ij}$ and $\widehat{a}_i \leftarrow \frac{n-1}{n-2}(\widehat{m}_i - T)$ for all i ▷ distinctiveness estimates

return T, $(\widehat{a}_i)_{i\in[n]}$, and the residuals $\widehat{r}_{ij} = d_{ij} - T - \widehat{a}_i - \widehat{a}_j$ for $\{i,j\} \in E$

Hence the error of G is unaffected by distinctiveness, for every table, if and only if G is balanced*, that is, weighted-regular ($\delta_i \equiv \bar{\delta}$). For unweighted designs this holds if and only if G is regular, that is, if every individual is compared equally often.*

4.2 Randomized balanced designs

A regular design removes distinctiveness, but a fixed regular graph can still align with relational structure, for instance by comparing only within roles. Randomizing the labels removes that risk.

Algorithm 1 is the simplest instance. It lays the relabeled individuals on a circle and compares each with its $\lfloor k/2 \rfloor$ nearest positions on either side; odd k (with n even) adds the $n/2$ diametral pairs. It belongs to the circulant partial diallel crosses of Kempthorne and Curnow (1961), here applied to diversity measurement; all relabeled circulant designs of the same degree have the same variance (Theorem 5). A second instance, used for error bars in Section 5.7, is the *round-robin design*[2]: after a uniform relabeling (for even n), the circle-method schedule partitions all N pairs into $n-1$ rounds of $n/2$ disjoint pairs, and the design keeps k rounds chosen uniformly at random. The following theorem covers every such construction.

Theorem 5 (Randomized balanced designs) *Let H be any simple k-regular graph on $[n]$ ($1 \le k \le n-1$, nk even), and let σ be a uniformly random permutation of $[n]$. Set $G = \sigma(H)$, $m = nk/2$, and $T = m^{-1}\sum_{e\in E(G)} d_e$. Then:*

(i) $T - \mu = m^{-1}\sum_{e\in E(G)} r_e$ *for every realization of σ;*

(ii) $\mathbb{E}[T] = \mu$;

(iii) $\mathrm{Var}(T) = \dfrac{\sigma^2_{\mathrm{int}}}{m} \cdot \dfrac{n-1-k}{n-3}$.

The mean and the variance of the error depend on H only through (n,k). In particular, $\mathrm{Var}(T)$ *does not depend on the expansion of H, and* $\mathrm{Var}(T) = 0$ *when $k = n-1$. Higher moments of the error can depend on H. All statements also hold when H is random and independent of σ, as for the round-robin design, since they hold conditionally on H.*

2. In the social relations model (Kenny and La Voie, 1984), a round-robin design observes every pair; here it means k rounds of a round-robin tournament schedule.

In practice, the variance is exact and a single round-robin design estimates it without bias (Proposition 15; Section 5.7); populations whose diversity is mostly level and distinctiveness are measured almost exactly; and once the labels are randomized, regularity, not expansion, determines the variance, so the simplest pattern is as accurate in mean square as a Ramanujan expander.

Corollary 6 (Efficiency over uniform pair sampling) *Let T_U be the mean of d over m pairs drawn uniformly without replacement from $\mathcal{P}$. Then*

$$\operatorname{Var}(T_U) \;=\; \frac{\sigma_{\text{tot}}^2}{m} \cdot \frac{N-m}{N-1}.$$

If $\sigma_{\text{int}}^2 > 0$, then for every $1 \le k \le n-2$ with nk even and $m = nk/2$,

$$\frac{\operatorname{Var}(T_U)}{\operatorname{Var}(T)} \;=\; (1+\kappa)\,\frac{n(n-3)}{(n-2)(n+1)}.$$

If $\sigma_{\text{int}}^2 = 0 < \sigma_{\text{tot}}^2$, the balanced design is exact while $\operatorname{Var}(T_U) > 0$. The ratio does not depend on k, and it tends to $1+\kappa$ as $n \to \infty$. The balanced design is therefore strictly more efficient if and only if $\kappa > 2(n-1)/\big(n(n-3)\big)$; otherwise it loses at most the factor $1 + 2(n-1)/\big(n(n-3)\big)$, which is 1.14 at $n = 16$ and 1.04 at $n = 50$.

The two variances have the same form when written per dimension of the relevant subspace; balance pays off exactly when distinctiveness carries more energy per dimension than relational structure does (Remark S5 in the Supplementary Material).

Theorem 13 in Appendix A.5 extends Theorem 5 to designs of arbitrary shape. Under a uniform relabeling, the variance depends only on the number of comparisons and the sum of squared degrees. When $\sigma_{\text{int}}^2 > 0$, $m < N$, and a regular design with m comparisons exists, regular designs are optimal exactly when $\kappa \ge 2(n-1)/\big(n(n-3)\big)$. Below that threshold, degree-concentrated designs do better, such as landmark designs, which compare every individual with a few random pivots (Section 5.1). The anatomy therefore indicates which kind of design to use.

The *efficiency factor* $1+\kappa$ is a property of the population alone: near 1 when individuals are equally distinctive, and large when a few individuals carry the diversity. The Horvitz–Thompson estimator under Bernoulli sampling is exposed to the raw second moment $N^{-1}\sum_{i<j} d_{ij}^2 = \mu^2 + \sigma_{\text{tot}}^2$ instead of σ_{tot}^2, so it is worse still whenever $\sigma_{\text{tot}}^2 \ll \mu^2$.

For i.i.d. observations, equireplicate designs are classical minimizers of the variance of incomplete U-statistics (Blom, 1976; Brown and Kildea, 1978; Lee, 1982, 1990); Remark S4 in the Supplementary Material relates them to the fixed-population results above.

4.3 A guarantee for every metric, with an optimal budget

Theorem 5 expresses the error through σ_{int}^2, which depends on the population. The triangle inequality bounds it in terms of the quantity being estimated.

Theorem 7 (A linear budget for every metric) *Every pseudometric table D satisfies*

$$\sigma_{\text{int}}^2 \;\le\; \mu^2 - \frac{2\,\sigma_{\text{tot}}^2}{n-2} \qquad \text{and hence, as } \sigma_{\text{int}}^2 \le \sigma_{\text{tot}}^2, \qquad \sigma_{\text{int}}^2 \;\le\; \frac{n-2}{n}\,\mu^2,$$

and both bounds are attained by the cut metric that splits an even population into two equal halves. Consequently, if $\mu > 0$, the design of Theorem 5 satisfies

$$\frac{\mathrm{Var}(T)}{\mu^2} \;\le\; v_{n,k} := \frac{2(n-2)(n-1-k)}{n^2(n-3)\,k} \;\le\; \frac{2}{nk},$$

and $\mathbb{P}\big(|T-\mu| > \varepsilon\mu\big) \le v_{n,k}/\varepsilon^2$ by Chebyshev's inequality. Given $\varepsilon > 0$ and $\eta \in (0,1)$, let k be the smallest degree with nk even and $k \ge \min\{n-1, \max\{1, 8/(n\varepsilon^2)\}\}$, and let $L = \lceil 8\ln(1/\eta)\rceil$. Lemma 16 then shows that the median of L independent such designs is a $(1\pm\varepsilon)$-approximation of μ with probability at least $1-\eta$. Each design uses at most $4\varepsilon^{-2}+n$ evaluations, so for every pseudometric table the total is

$$O\big((n+\varepsilon^{-2})(1+\log(1/\eta))\big).$$

The first bound is equivalent to $\sum_i S_i^2 \ge n\sum_{i<j} d_{ij}^2$, which follows from the averaged triangle inequality $d_{ij} \le (S_i+S_j)/n$; for Euclidean data it is the inequality $T_{1,n} \le 2T_{3,n}$ that Edelmann et al. (2020, Lemma 4.1) use to show that the distance standard deviation is at most Gini's mean difference, with the same two-point equality case. In distance-covariance terms, $\sigma^2_{\mathrm{int}} = \frac{n-3}{n-1}\Omega_n$ (Section 3). Its use here is to bound the variance of every balanced design: since $v_{n,k} \le 2/(nk) = 1/m$, a balanced design with m comparisons has relative root-mean-square error at most $m^{-1/2}$ for every pseudometric table.

A single balanced design also concentrates exponentially, but bounded-difference bounds depend on the range $d_{\max}/\mu$, which can grow linearly in n over pseudometrics; Chebyshev's inequality and a median avoid this, so the guarantee is scale-free (Remark S6 in the Supplementary Material).

Two consequences are worth stating. With $k = 1$ (n even), a single uniformly random perfect matching of $n/2$ pairs has $\mathrm{Var}(T)/\mu^2 \le v_{n,1} \le 2/n$, so it estimates μ within a factor $1\pm\sqrt{8/n}$ with probability at least $3/4$ for every pseudometric table; Chang (2017) analyzed the same design for one-sided constant-factor estimates. For bounded dissimilarities that need not be metrics, $0 \le d_{ij} \le B$, as judge scores are, the elementary bound $\sigma^2_{\mathrm{int}} \le \sigma^2_{\mathrm{tot}} \le \mu(B-\mu)$ gives $\mathrm{Var}(T)/\mu^2 \le (B/\mu-1)(n-1-k)/\big(m(n-3)\big)$ without the triangle inequality.

The budget has two terms. The n term pays for comparing every individual at least once. The ε^{-2} term is the usual price of estimating a mean. Neither can be avoided. Uniform pair sampling can need about $n/2$ times more comparisons: setting $\sigma^2_{\mathrm{int}} \ge 0$ in Theorem 7 gives $\sigma^2_{\mathrm{tot}} \le (n-2)\mu^2/2$, attained by the star metric (one individual at distance 1 from $n-1$ coincident others), on which every balanced design is exact.

Proposition 8 (The linear budget is optimal) *Up to constant factors, and for constant failure probability, the budget of Theorem 7 is optimal; balanced designs attain it non-adaptively, and uniform pair sampling does not.*

(i) For every $n \ge 4$, any (possibly adaptive and randomized) procedure that evaluates fewer than $n/8$ entries fails, with probability at least $3/8$ on some pseudometric table, to return $\widehat{\mu}$ with $\widehat{\mu}/\mu \in (3^{-1/2}, 3^{1/2})$.

(ii) There is a universal $c > 0$ such that, for every $\varepsilon \in (0, 1/8]$ and every n with $N \ge 4096\,\varepsilon^{-2}$, any procedure that returns a $(1\pm\varepsilon/2)$-approximation of μ with probability at least $2/3$ on every pseudometric table with entries in $[1,2]$ must evaluate at least $c\,\varepsilon^{-2}$ entries on some such table.

(iii) Uniform pair sampling does not reach this budget (compare Barhum et al., 2007, Proposition 2). On the one-outlier tables $D^{(u)}$ ($d_{uj} = 1 + n$ for $j \neq u$, all other off-diagonal entries 1), the mean T'_U of m pairs drawn uniformly with replacement has $\mathrm{Var}(T'_U)/\mu^2 = 2(n-2)/(9m)$. It therefore needs $m = \Theta(n/\varepsilon^2)$ pairs for relative root-mean-square error ε. Every balanced design is exact on $D^{(u)}$, because $R = 0$.

Together, (i) and (ii) show that $\Omega(n + \varepsilon^{-2})$ evaluations are necessary whenever $\varepsilon \leq 1/8$ and $N \geq 4096\,\varepsilon^{-2}$. This matches Theorem 7 up to constants and the $\log(1/\eta)$ factor.

Adaptive schemes with unequal weights can also reach linear budgets. Chechik et al. (2015) compute the n distances to one well-chosen pivot, use them as constant-factor proxies for every individual's total distance, and then sample $O(\varepsilon^{-2})$ pairs with probabilities proportional to these proxies. Balanced designs reach the same budget non-adaptively: all pairs are fixed before any comparison is made, the weights are uniform, the variance is known in closed form for every degree and design shape, and all comparisons can be evaluated in one parallel batch.

4.4 Fixed designs: expanders and local graphs

Some applications need a fixed comparison graph: a communication topology, a graph that must be shared across training iterations so that changes in the estimate are not sampling noise, or a deterministic certificate. For fixed regular graphs, the error is controlled by how well the graph mixes.

Theorem 9 (Deterministic guarantee for fixed regular designs) *Let G be a fixed simple k-regular graph with adjacency eigenvalues $k = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.*

(i) For every dissimilarity table D,

$$|\widehat{\mu}_G - \mu| \;\leq\; \frac{(\lambda_2 - \lambda_n)\,\|R\|_*}{2nk},$$

or equivalently, if $\mu > 0$, $|\widehat{\mu}_G/\mu - 1| \leq (\lambda_2 - \lambda_n)\,\|R\|_*/(2nk\mu)$, where $\|\cdot\|_*$ is the nuclear norm. If D is a pseudometric table, then $\|R\|_*/(n\mu) \leq \sqrt{\mathrm{rank}(R)\,(n-1)(n-2)}/n$.

(ii) For every diagonal matrix Λ,

$$|\widehat{\mu}_G - \mu| \;\leq\; \frac{\lambda}{nk}\,\|P(R + \Lambda)P\|_* + \frac{|\mathrm{tr}\,\Lambda|}{n^2},$$

where $\lambda = \max(|\lambda_2|, |\lambda_n|)$ and $P = I - \mathbf{1}\mathbf{1}^\top/n$.

Random k-regular graphs with fixed $k \geq 3$ satisfy $\lambda_2 - \lambda_n \leq 4\sqrt{k-1} + o(1)$ with high probability as $n \to \infty$ (Friedman, 2008; Bordenave, 2020).

The bound depends on the relational part only, because regularity cancels distinctiveness and the level exactly. Since $\lambda_2 \geq 2\sqrt{k-1} - o(1)$ for every k-regular graph (the Alon–Boppana bound; Hoory et al., 2006), the bound is informative only when $\|R\|_*$ is small compared with $n\mu\sqrt{k}$: when R is close to low rank, possibly up to a diagonal correction (part (ii) and Remark 17), or when $\sigma_{\mathrm{int}}/\mu$ is small. No fixed sparse design can be accurate

uniformly over metrics (Remark 18), so this guarantee is necessarily instance-dependent, through $\|R\|_*$; randomization is what removes the dependence. For Euclidean and, more generally, ℓ_1-embeddable tables, Barhum et al. (2007) show that the average over a k-regular graph whose nontrivial eigenvalues are at most λ in absolute value is within a factor $1 \pm 2\lambda/k$ of μ, so Ramanujan graphs of degree $O(\varepsilon^{-2})$ are deterministic $(1 \pm \varepsilon)$-approximators. This covers SND with Gaussian policies of diagonal covariance, whose W_2 is a Euclidean distance between (mean, standard deviation) vectors; averages of Euclidean distances embed isometrically in ℓ_1. Part (ii) handles role structure. For a population with K roles and role-level table F (Section 3), choosing $\Lambda = \mathrm{Diag}(F)$ makes $P(R+\Lambda)P$ a matrix of rank at most K. An expander's error is then $O(\sqrt{K/k})\,\sigma_{\mathrm{int}}/\mu + O(1/n)$ in relative terms (when every role has at least two members and $\max_i |F_{ii}| = O(\mu)$; Remark 17), uniformly over all labelings, including adversarial ones. A fixed graph with poor expansion, such as a ring of nearest neighbors, has $\lambda_2 \approx k$ and no such protection. If its neighborhoods coincide with roles, it misses the between-role diversity entirely.

Local graphs are therefore best understood as defining a different quantity, the diversity among the pairs they connect. That quantity is useful when interactions are what matter, for example when only communicating agents need to differ. Proposition 3 gives its exact relation to the global μ: the gap is the distinctiveness leakage plus the average residual on the local edges. Both terms can be estimated from a small balanced design run alongside, so a practitioner can report both quantities and their difference.

4.5 Estimating distinctiveness from the same comparisons

The comparisons made by a balanced design also estimate who is distinctive.

Proposition 10 (Sparse distinctiveness) *Under the design of Theorem 5, the comparison partners of each individual i form a uniformly random subset of size k of the other individuals. Let $\widehat{m}_i$ be the average of the k observed dissimilarities of i, and let $\widehat{a}_i = \frac{n-1}{n-2}\,(\widehat{m}_i - T)$. Then $\mathbb{E}[\widehat{a}_i] = a_i$ for every i, and*

$$\mathrm{Var}(\widehat{m}_i) = \frac{n-1-k}{n-2} \cdot \frac{s_i^2}{k}, \qquad s_i^2 = \frac{1}{n-1} \sum_{j \neq i} \Big(d_{ij} - \frac{S_i}{n-1} \Big)^2,$$

and $\mathrm{Var}(\widehat{a}_i) \le 2\big(\frac{n-1}{n-2}\big)^2 \big[\mathrm{Var}(\widehat{m}_i) + \mathrm{Var}(T)\big]$, *which is* $O\big((s_i^2 + \sigma_{\mathrm{int}}^2/n)/k\big)$.

With k comparisons per individual, the standard error of $\widehat{a}_i$ is therefore about $s_i/\sqrt{k}$ when $s_i^2 \gtrsim \sigma_{\mathrm{int}}^2/n$, which is enough to rank individuals and flag outliers (Section 5.7). Recovering role structure from the observed residuals is a matrix-completion problem for the low-rank table F on a random regular sampling graph (Candès and Recht, 2009), which we leave to future work. Table 1 summarizes what each design estimates and how well.

4.6 From measurement to control

Diversity can also be a control variable rather than a diagnostic. DiCo (Bettini et al., 2024b) gives each agent a shared component and an agent-specific component, measures the diversity of the specific components, and rescales them so that the team's SND equals a target

Table 1: **What each comparison design estimates, and how well.** Here m is the number of comparisons, "local div." is the mean dissimilarity over the graph's edges (its bias for μ is given by (7)), $v_{n,k}$ is defined in Theorem 7, and HT is the Horvitz–Thompson estimator. "Leaks a" means the error contains distinctiveness leakage (Proposition 3). *The bound holds for pseudometric tables (Theorem 7).

Design	Target	Bias for μ	Leaks a	Variance or bound	Source
Complete graph ($m = N$)	μ	0	no	0	—
Uniform pairs	μ	0	yes	$\frac{\sigma^2_{\text{tot}}}{m}\frac{N-m}{N-1}$	Corollary 6
Bernoulli (HT)	μ	0	yes	$\frac{1-p}{m}(\mu^2+\sigma^2_{\text{tot}})$, $m = pN$	Remark 19
Agent subsampling (s agents)	μ	0	strongly	$\frac{2(n-s)}{(n-2)s}\sigma^2_{\text{main}} + O(\sigma^2_{\text{int}}/m)$	Theorem 13
Randomized balanced	μ	0	no	$\frac{\sigma^2_{\text{int}}}{m}\frac{n-1-k}{n-3}$ ($\le v_{n,k}\,\mu^2$)*	Theorem 5
Fixed expander	μ	$\le \frac{\lambda_2-\lambda_n}{2nk}\|R\|_*$	no	0 (deterministic)	Theorem 9
Local graph	local div.	(7)	yes	—	Proposition 3

μ_{des}. The reference implementation recomputes the metric on every training minibatch, freezes it during rollouts, and smooths it, $\widehat{\mu} \leftarrow (1-\tau)\widehat{\mu} + \tau\,\mu_{\text{batch}}$ ($\tau = 0.01$ by default), so the quadratic cost of exact evaluation is paid at every gradient step. Because the rescaling is multiplicative, an estimation error turns into a tracking error in a simple way.

Proposition 11 (Tracking guarantee for diversity control) *In a controller of the DiCo type, each agent i acts with mean $b(o) + c\,h_i(o)$, where b is shared and h_i is agent-specific. The scale is $c = \mu_{\text{des}}/\widehat{\mu}$, where $\widehat{\mu} > 0$ estimates the diversity $\mu(h) > 0$ of the unscaled components. Assume that the behavioral distance is positively homogeneous of degree one in the agent-specific components, as W_2 distances between Gaussians with a shared covariance are. Let $e = \widehat{\mu}/\mu(h) - 1$. Then the realized diversity is $\mu_{\text{real}} = c\,\mu(h) = \mu_{\text{des}}/(1+e)$, and:*

(i) if $|e| \le \varepsilon < 1$, then $|\mu_{\text{real}}/\mu_{\text{des}} - 1| \le \varepsilon/(1-\varepsilon)$;

(ii) if $\widehat{\mu}$ is the balanced estimator T of Theorem 5 and the distances form a pseudometric table, then $\mathbb{E}[1/\mu_{\text{real}}] = 1/\mu_{\text{des}}$ exactly (with the convention $1/\mu_{\text{real}} := 0$ on the event $\{T = 0\}$, which is empty when all off-diagonal distances are positive); moreover, for every $0 < \varepsilon < 1$,

$$\mathbb{P}\Big(\Big|\frac{\mu_{\text{real}}}{\mu_{\text{des}}} - 1\Big| > \frac{\varepsilon}{1-\varepsilon}\Big) \le \frac{v_{n,k}}{\varepsilon^2},$$

and the tracking bias is of second order,

$$\big|\mathbb{E}\big[(\mu_{\text{real}}/\mu_{\text{des}} - 1)\,\mathbf{1}\{|e| \le 1/2\}\big]\big| \le 4\,\mathbb{E}[e^2] \le 4v_{n,k},$$

while $\mathbb{P}(|e| > 1/2) \le 4v_{n,k}$.

In words, to first order the controller inherits the estimator's relative error one for one, and an unbiased estimator yields a tracking bias of order $\mathbb{E}[e^2]$, about 10^{-4} for a per-call relative error of 1%. Averaging over iterations shrinks the fluctuating part of the tracking error, because each design is drawn independently of the past; the second-order bias does

not average out. Under the soft update, independent per-call errors of variance ν leave variance $\tau\nu/(2-\tau)$, at the price of lag. Our simulation in Section 5.8 uses $\tau = 1$, the hardest case. Proposition 11 is a per-step statement for a fixed population; it does not model how policies adapt to the rescaling during learning, which we study empirically in Section 5.8.

5 Experiments

The experiments answer seven questions. Do the closed-form error laws hold on synthetic and real tables (Section 5.2)? How do comparison designs rank at equal cost, and when does each one fail (Section 5.3)? What does the anatomy reveal about trained populations, and what does it imply for measuring them (Sections 5.4 and 5.5)? Do the gains carry over to expensive comparisons that are not metrics, such as neural judges of generated text (Section 5.6)? Do the same comparisons give valid error bars (Section 5.7)? Does a sparse estimate suffice when diversity is a control signal (Section 5.8)? And what does it cost at scale (Section 5.9)? Unless stated otherwise, every error is the relative root-mean-square error (RMSE) over design draws, that is, the root-mean-square of $\widehat{\mu}/\mu - 1$ against the exact all-pairs value. Every Monte Carlo cell in Section 5.2 uses 3,000 independent design draws on a fixed table (the judge cells of Section 5.6 use 4,000, and other experiments 2,000 unless stated). With 3,000 draws, a ratio of a measured to a predicted variance has a Monte Carlo standard error of about $\sqrt{2/2999} \approx 2.6\%$, and a ratio of two measured variances about 3.7%. Full configurations are in Section S2 of the Supplementary Material, and all code, tables, and logs are released.

5.1 Populations and designs

Populations. *Synthetic families* isolate one anatomical component each. They include a near-simplex (all individuals nearly equidistant), a one-outlier population, a heterogeneous-scale population (individual i is a Gaussian point in $\mathbb{R}^{64}$ scaled by a log-normal factor), two equal roles, four unequal roles (40/30/20/10%), and a continuum (individuals spread on a circle). Each family is drawn at $n \in \{50, 100, 200\}$ with two seeds for the law checks, plus three further tables at $n = 200$ for the equal-cost comparison; all synthetic tables use Euclidean distance. *Trained MARL teams* are teams of independently parameterized Gaussian policies trained with independent PPO on VMAS tasks (Bettini et al., 2024a). We record the full SND distance table (4) at initialization and at several points during training, and form *reset variants* of trained teams by resetting one agent, or half of the agents, to their initial weights. *Model populations* are 128 or 1,024 small neural classifiers that differ in seed and training recipe, compared by the mean Jensen–Shannon distance of their predictions on a held-out set. *Generated text* consists of populations of 64, 300, or 1,000 responses from Qwen2.5-0.5B-Instruct, Llama-3.2-1B-Instruct, or Claude Opus 5.5, compared by a cross-encoder judge, by Claude Opus 5.5, or (on one table) by GPT-5. *Image sample sets* contain up to 60,000 Fashion-MNIST images (Xiao et al., 2017), compared by the Euclidean distance between classifier features. *Closed-loop runs* are simulated DiCo-style controllers on drifting populations of $n = 50$ or 500 agents, and DiCo diversity-control runs on VMAS Dispersion with $n = 50$ agents.

Designs. We compare the randomized balanced design of Algorithm 1 (permuted circulant) and its round-robin version against uniform pair sampling without replacement, Bernoulli sampling with the Horvitz–Thompson and ratio estimators, agent subsampling (all pairs among a random subset), a landmark design that compares every individual with a few random pivots, a k-nearest-neighbor graph built from the table itself, a fixed ring lattice, and a fixed random k-regular expander. Budgets are matched to $m = nk/2$ comparisons, in expectation for Bernoulli designs; landmarks at $k = 1$ (one pivot, $n - 1 \approx 2m$ pairs) and symmetrized nearest-neighbor graphs use up to about $2m$, which favors them.

5.2 Checking the closed-form laws

Figure 2(a) compares measured and predicted design variances over 1,576 cells. They cover 36 synthetic tables (six families, $n \in \{50, 100, 200\}$, two seeds) and 67 real tables: 66 VMAS tables (63 from the first nine training runs, including their reset variants, and three from independently initialized teams) and the model population of Section 5.5; each cell combines one table, one of four designs, and one degree k. On the synthetic tables, the ratio of measured to predicted variance (mean $\pm$ standard deviation over cells) is 1.001 ± 0.026 for the round-robin design, 0.996 ± 0.025 for the permuted circulant, 1.003 ± 0.024 for uniform sampling, and 1.000 ± 0.024 for Bernoulli sampling with the Horvitz–Thompson estimator. On the real tables the ratios are 0.998 ± 0.026, 1.001 ± 0.026, 1.000 ± 0.023, and 1.003 ± 0.029, respectively. These spreads are close to the Monte Carlo standard error of a variance estimated from 3,000 draws (0.026), and 94.9% of all cells lie within two standard errors of the prediction (Figure 2(a)), so the measurements agree with the closed-form laws of Theorem 5, Corollary 6, and Remark 19 to within Monte Carlo error. Because these laws are exact identities, this check verifies the derivations and the implementation; the empirical question is how large κ is in real populations, which Sections 5.4 to 5.6 answer. The variances of the circulant and round-robin designs agree with each other, confirming that once labels are randomized the pattern does not affect the variance.

Figure 2(b) tests the efficiency law of Corollary 6, which predicts the variance ratio between uniform and balanced designs from the anatomy alone. Medians of measured versus predicted ratios are 2.04 versus 2.07 for the near-simplex family, 45.7 versus 45.9 for heterogeneous scale, 225 versus 229 for one outlier, 1.02 versus 1.03 for two equal roles, 1.18 versus 1.20 for four unequal roles, and 0.98 versus 0.98 for the continuum. The last value is the slight $O(1/n)$ advantage of uniform sampling that the corollary predicts when $\kappa \approx 0$. The efficiency factor therefore ranges over two orders of magnitude, and it is predicted to within Monte Carlo error by a quantity computed from the table. It falls below one only when $\kappa < 2(n-1)/\big(n(n-3)\big)$, and then only by the small factor that Corollary 6 allows.

5.3 Designs at equal cost

At $n = 200$, Table 2 compares ten designs on the six synthetic families, which isolate the failure modes predicted by Proposition 3, at a budget of $m = 2n$ comparisons (2% of all pairs). Five findings stand out, and each is a direct consequence of the anatomy.

- *Balanced designs are the most reliable choice.* Here "balanced" means the randomized balanced design or, equivalently on average, a random regular graph redrawn for each

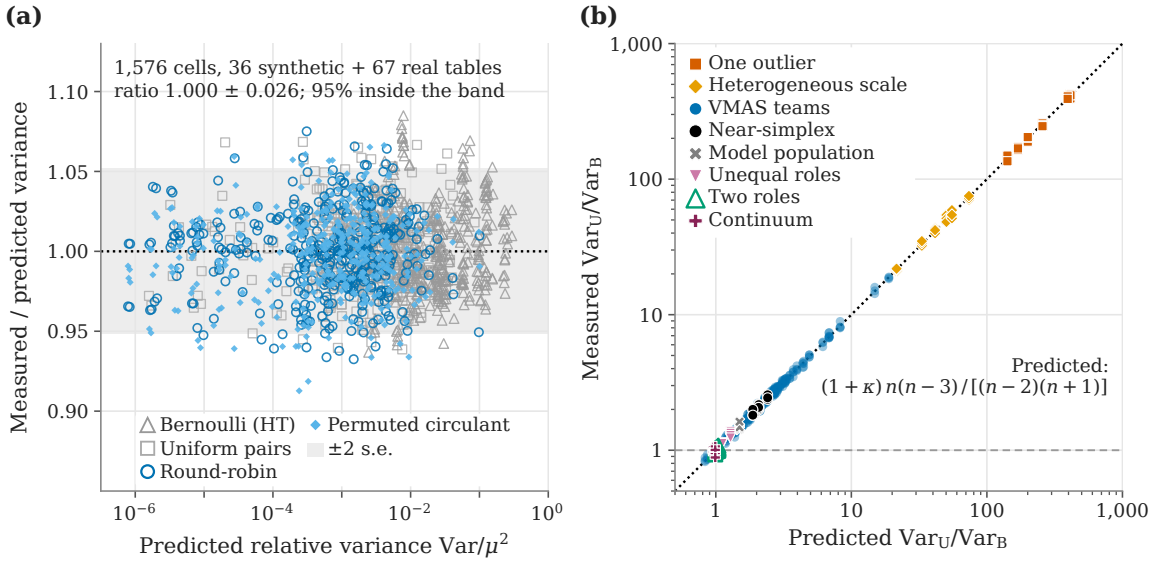


Figure 2: **Implementation check: measured variances match the exact closed forms.** (a) Ratio of measured to predicted variance for four designs, against the predicted relative variance, over 1,576 cells: 36 synthetic tables and 67 real tables (VMAS teams, including reset variants, and the model population), several degrees k, and 3,000 draws per cell. Round-robin and permuted circulant are the two balanced designs. The band is ±2 Monte Carlo standard errors of a variance estimated from 3,000 draws; 95% of the cells fall inside it, as expected if the laws are exact. (b) Measured versus predicted efficiency of balanced over uniform designs (Corollary 6), from near 1 for relational populations to about 140–420 for a single outlier. $\mathrm{Var_U}$ and $\mathrm{Var_B}$ are the variances of the uniform and balanced estimates at equal m; the diagonal is $y = x$, and the dashed line marks equal efficiency.

estimate. Its error is the smallest, or within 1.2× of the smallest, in five of the six families. Where individuals differ in distinctiveness, it beats uniform sampling by 5.7–12.8× in error (ratios of unrounded values), that is, by 32–163× in variance at equal cost. Where diversity is mostly relational, it nearly ties uniform sampling, as Corollary 6 predicts. The one exception is the continuum, where every individual has the same average distance and comparing everyone with a few pivots is more accurate. This exception is predicted by Theorem 13: the continuum's $\kappa = 0.0008$ lies below the threshold $2(n-1)/\big(n(n-3)\big) = 0.010$, below which designs with uneven degrees have smaller variance.

- *Horvitz–Thompson weighting, one of the Bernoulli estimators of Graph-SND* (Ray, 2026)*, is uniformly poor* (4.9–5.7%). Its variance carries μ^2 in addition to σ^2_{tot}, so it cannot exploit any structure. The ratio form of the same design behaves like uniform sampling.

Table 2: **Relative error (RMSE, %) of ten designs at equal cost.** Synthetic populations with $n = 200$; every design uses $m = 2n = 400$ (agent subsampling $\binom{29}{2} = 406$, landmarks 397, Bernoulli designs in expectation; the adaptive pivot design of Chechik et al. (2015), CCK, spends $n-1$ comparisons on a random pivot and samples $n+1$ pairs; symmetrized nearest-neighbor graphs up to $2m$) of the 19,900 pairs (2,000 draws per cell, three tables per family; the fixed expander is averaged over 100 independently drawn graphs). Rows marked † are fixed structured graphs (the ring follows the generator's ordering, by role or by angle for the last three families; the nearest-neighbor graph is built from the table itself); their error is pure bias, one realization per table; on the first three, exchangeable families the ring's expected error equals the balanced design's (Theorem 5). "Balanced" is the round-robin design; HT is the Horvitz–Thompson estimator; "unequal roles" has four roles of sizes 40/30/20/10%. Bold: within 5% of the best randomized or averaged entry in the column (Monte Carlo errors below 2%).

Design	near-simplex	het. scale	one outlier	two roles	unequal roles	continuum
Balanced	**0.13**	0.38	**0.13**	**2.45**	**2.75**	2.37
Random expander (fixed)	**0.13**	**0.36**	**0.13**	**2.55**	**2.82**	2.42
Uniform pairs	0.18	2.14	1.61	**2.49**	2.91	2.34
Bernoulli (ratio)	0.18	2.19	1.59	**2.47**	2.93	2.39
Bernoulli (HT)	4.93	5.36	5.17	5.48	5.74	5.49
Agent subsampling	0.64	10	7.89	3.34	6.26	2.05
Landmarks	1.29	21	15	5.34	12	**0.68**
Adaptive pivot (CCK)	0.24	2.20	1.27	3.49	4.10	3.53
Nearest neighbors†	7.28	33	8.39	69	90	96
Fixed ring†	0.15	0.32	0.14	46	81	96

- *Subsets are expensive.* Agent subsampling, which measures a sub-team, and landmark designs, which compare everyone to a few pivots, concentrate comparisons on few individuals and pay for distinctiveness many times over. They are 5–122× worse in error than balanced designs unless diversity is mostly relational.
- *Adaptivity does not pay at this budget.* The adaptive estimator of Chechik et al. (2015) samples pairs with probabilities proportional to distance proxies from one pivot. It is unbiased but 1.4–10× less accurate than the balanced design in every family, and on the real tables of Sections 5.5 and 5.6 and the VMAS teams it is 1.4–2.8× less accurate than the balanced design and less accurate than uniform pairs, because the pivot uses half of the budget.
- *Fixed structured graphs are unbiased only by luck.* A nearest-neighbor graph underreports diversity by 7–96%, because it compares the most similar pairs, which have negative residuals and over-represent the least distinctive individuals, so it suffers from both leakage terms of Proposition 3. A fixed ring is accurate when labels carry no information. When labels are ordered by role or by angle, as they often are in practice, it misses 46–96% of the diversity (46–81% for role orderings). Randomizing the labels is what makes the guarantee of Theorem 5 unconditional. A random regular graph that is drawn once and then fixed matches the randomized design on average

over draws, as Theorem 5 implies (the union-of-matchings expander of Table 2 keeps repeated edges, which raises its mean squared error by less than 9%; Section S2); its additional value is the worst-case guarantee of Theorem 9. That bound held in every case we tested (six families with $k \in \{4, 8, 16\}$ and the 1,024-model population with $k \in \{4, 8\}$), and it is nearly attained by rings aligned with roles or angles, whose errors come within 1.2–2.3× of it, while random expanders sit at least 18× below it (18–135× for the closest of the 20 graphs drawn per setting, 58–310× on average).

Figure S1 in Section S3 shows the same qualitative picture at five budgets from $m = n/2$ to $m = 8n$. With $m = n/2$ (one comparison per individual, 0.5% of all pairs), the one-outlier population is measured with 0.25% error by the balanced design, against 3.2% for uniform pairs, 11.5% for agent subsampling, and 23.5% for landmarks with a single pivot, which already uses $n - 1 = 199$ comparisons.

5.4 The anatomy of trained populations

We trained teams of $n \in \{8, 16, 32\}$ agents with independent PPO on two VMAS tasks with different structure: Navigation, in which every agent must reach its own goal, and Dispersion, in which all agents spawn at the center and must consume n randomly placed food particles, each once (details in Section S2.3). Runs stop after 10–205 iterations (0.08–1.7 million environment steps) under fixed wall-clock budgets, so they show teams early in training; seven longer runs reach up to 766 iterations (below). Figure 3 shows two of the recorded tables next to two synthetic references, and Figure 4 summarizes all of them.

Diversity grows, and changes form. Independently initialized policies are almost interchangeable: μ is 0.003–0.006 at initialization, and distinctiveness carries 31–69% of the pairwise variance. Training raises μ by more than an order of magnitude, to 0.075–0.11 in Navigation and 0.21–0.24 in Dispersion, but the anatomy shows that the two tasks get there differently. In all eight Navigation runs with $n \in \{16, 32\}$, distinctiveness carries 49–88% of the variance at every recorded iteration from iteration 5 on and ends within 13 points of its maximum. In all six Dispersion runs it first rises to 63–87% at iterations 5–10 and then falls by 23–62 points, to 17–40%, which separates the tasks without overlap (exact two-sided permutation $p = 2/\binom{14}{6} \approx 7 \times 10^{-4}$, exploratory because we chose the statistic after seeing the trajectories; the two noisier $n = 8$ Navigation runs fall by 26 and 30 points, and with them $p = 0.001$). Seven longer $n = 16$ runs (bold in Figure 4; Section S2.3) remove the difference in run length. At iteration 300, the share is 6–7% in three Dispersion teams and 75–81% in two Navigation teams; at the end (543–766 iterations, up to 6.3 million steps) it is 2–3% and 58–64%, against 28–70% for unstructured populations of independent Gaussian points (5th to 95th percentiles). Two Dispersion teams with a centralized critic, as in MAPPO (Yu et al., 2022), fall more slowly, from 76–82% to 7–28% after 348–535 iterations. SND alone reports only that diversity increased. We read the contrast as a hypothesis: in Dispersion all agents spawn at one point with identical observations, so a good team must act differently on the same input, whereas one goal-conditioned policy solves Navigation; functional heterogeneity would then appear as relational structure and incidental heterogeneity as distinctiveness. Confounds remain: the observation pool changes between iterations, and each agent's learned action standard deviation enters W_2.

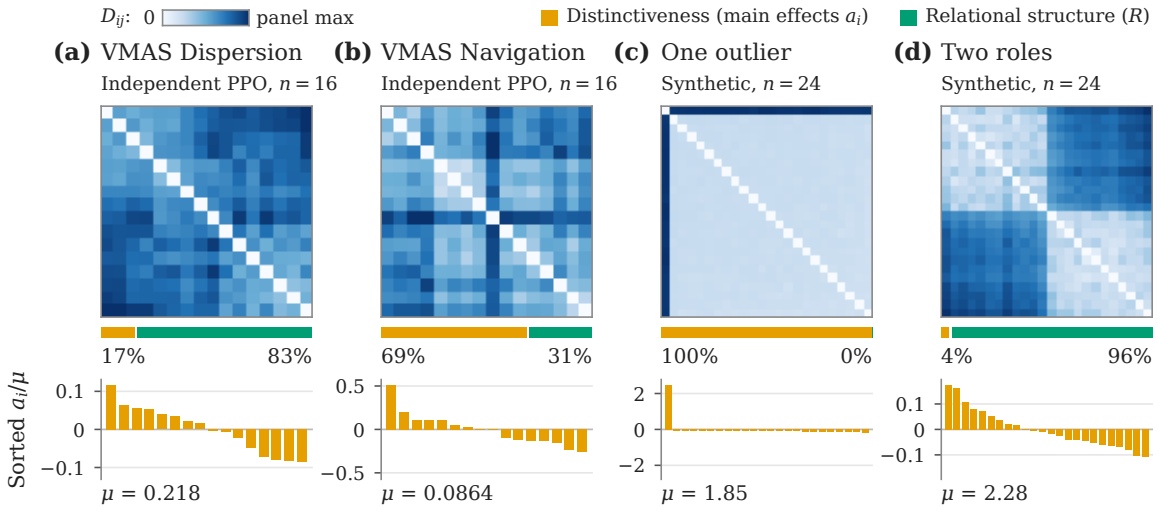


Figure 3: **The anatomy of diversity in four populations.** (a, b) SND behavioral-distance tables of heterogeneous teams ($n = 16$) after 41 iterations of independent PPO on VMAS Dispersion and 35 iterations on VMAS Navigation (early in training). (c, d) Synthetic populations generated as in Table S2 with $n = 24$. Top: the dissimilarity table D, with individuals ordered by the leading eigenvector of R when R carries at least a quarter of the pairwise variance (a, b, d), and by distinctiveness a_i otherwise (c). Each panel has its own color scale, from 0 (white) to the panel's largest entry (key at top left), so darker means more dissimilar. Middle: shares of the pairwise variance carried by distinctiveness ($\sigma^2_{\text{main}}/\sigma^2_{\text{tot}}$, orange) and by relational structure ($\sigma^2_{\text{int}}/\sigma^2_{\text{tot}}$, green), as in Theorem 2. Bottom: sorted distinctiveness a_i/μ, with the level μ printed below each panel. The Dispersion team (a) is dominated by relational structure, which varies along a single gradient rather than in blocks. The Navigation team (b) is dominated by individual distinctiveness. Balanced designs are exact on the orange part, and their error is governed by the green part alone (Theorem 5).

Consequences for measurement. The predicted variance ratio of uniform to balanced designs (Corollary 6) ranges from 1.05 to 7.1 on these teams (an RMSE gain of 1.02–2.7×), with median 2.4 (interquartile range 1.8–3.2) over the 70 tables recorded after initialization: a balanced design typically halves the comparisons needed for a given accuracy, and the much larger factors of Table 2 arise only when a few individuals dominate distinctiveness. The ratio is largest when a few agents have drifted far from the rest and smallest when diversity is mostly relational; in the longer runs it reaches 7.2 in Navigation but falls slightly below one (0.89–0.99) in late Dispersion teams, whose diversity is almost purely relational, as Corollary 6 allows. For example, with $m = 2n$ comparisons on one Navigation team with $n = 32$, the measured relative error is 2.16% for the balanced design and 3.95% for uniform pairs (predicted 2.17% and 3.90%), while on a late Dispersion team the two coincide (3.01% and 3.05%). When one agent of a team is reset to its initialization, which mimics a failed or

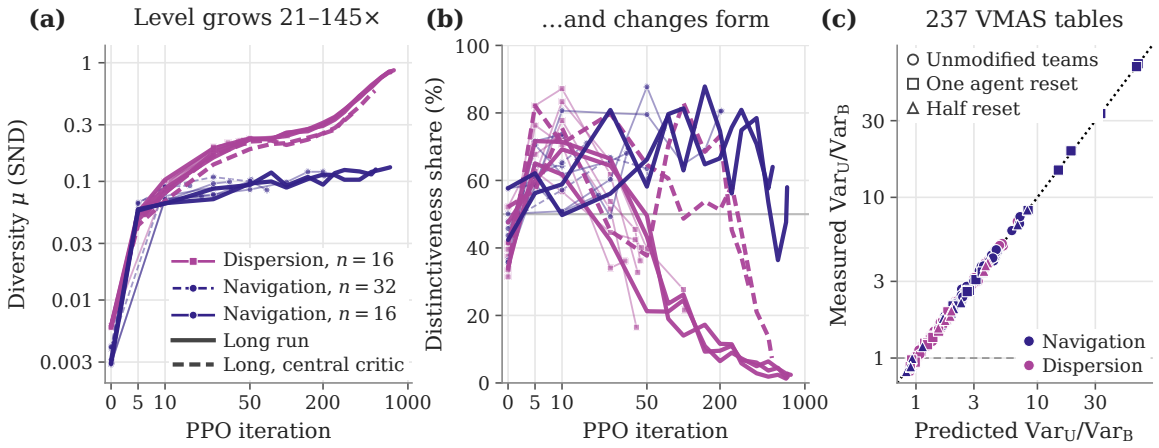


Figure 4: **The anatomy of multi-agent teams during training.** Independent PPO teams on VMAS Navigation with $n = 16$ (five seeds) and $n = 32$ (three seeds) and on VMAS Dispersion with $n = 16$ (six seeds); one line per seed (horizontal axis linear up to iteration 10 and logarithmic beyond), each run as long as its compute budget allowed; bold: the seven longer $n = 16$ runs (dashed: centralized critic). (a) Diversity μ (SND) grows by more than an order of magnitude from initialization. (b) Share of the pairwise variance carried by distinctiveness (gray line: 50%); the rest is relational. In Dispersion, which rewards agents for covering different targets, diversity first concentrates in individuals and then moves toward relational structure. (c) Measured ratio of uniform to balanced variance ($k = 4$; 4,000 draws each) against the prediction of Corollary 6 for all 237 VMAS tables of all runs, including the two $n = 8$ Navigation runs, an $n = 64$ initialization table, and teams with one agent or half of the agents reset to initialization; the median ratio of measured to predicted is 1.00, and almost purely relational tables have predicted ratios slightly below 1 (0.83–0.99), as Corollary 6 allows. Color indicates the task as in (a, b); dotted: $y = x$; dashed: equal efficiency.

newly added teammate, the ratio ranges from 1.2 to 33 over the 16 teams (median 2.7; 65–68 for the two longer Navigation runs); it is largest for the Navigation teams trained longest and smallest for Dispersion teams (1.2–1.7, and 0.90–1.8 for the longer runs). Resetting half of a 32-agent team, which gives experts and novices, gives 2.1–8.3; for three half-reset Navigation teams, however, κ is only 0.06–0.17 and the ratio is 0.83–0.93 (Figure 4(c)), again as Corollary 6 allows.

5.5 A population of trained models

The theory applies to any population with a pairwise dissimilarity. As a second domain we trained 128 small multilayer perceptrons on Fashion-MNIST with four training recipes (32 seeds each): a narrow network trained for one epoch, a wide network trained for three epochs, the wide network with 30% label noise, and a medium network with input-noise

augmentation. Their mean test accuracies are 0.81, 0.87, 0.87, and 0.86, respectively. The dissimilarity of two models is the Jensen–Shannon distance between their predictive distributions, averaged over 2,000 test inputs. This distance is a pseudometric but is not separable in the sense of Proposition S1, so all-pairs evaluation costs 8,128 model-pair comparisons. The anatomy of this population is $\mu = 0.229$ with a coefficient of variation of 0.51; distinctiveness carries 34% of the pairwise variance and relational structure 66%. The relational part is almost exactly low-rank: its leading eigenvector carries 86% of the relational energy and the leading three carry 98%. Three is the rank that four groups produce once main effects are removed. Clustering the models by the three leading eigenvectors of R (k-means with four clusters) recovers the four training recipes perfectly (adjusted Rand index 1.0) from the all-pairs dissimilarity table alone, without access to the recipe labels, and the distinctiveness profile singles out the label-noise recipe as the most distinctive (mean $a_i = +0.080$). With $m = 2n = 256$ of the 8,128 pairs, the measured relative error is 2.56% for the balanced design (predicted 2.56%), 3.11% for uniform pairs (predicted 3.13%), and 6.98% for Bernoulli sampling with the Horvitz–Thompson estimator (predicted 6.90%).

The same experiment at eight times the size gives the same picture. A population of $n = 1{,}024$ models (256 seeds per recipe, trained in under two minutes on one GPU) has 523,776 model pairs, and its anatomy is the same as before: $\mu = 0.227$, 34% distinctiveness, 66% relational, with the leading three eigenvectors of R carrying 99.3% of the relational energy. With eight comparisons per model (4,096 of the 523,776 pairs), the balanced design estimates μ to a relative error of 0.62%, against 0.78% for uniform pairs. At the same cost, agent subsampling, landmarks, and a nearest-neighbor graph err by 4.2%, 10.4%, and 65%, respectively. Measured variances agree with the closed-form predictions to within Monte Carlo error at every degree tested.

5.6 Neural judges and sample sets at scale

Some of the most expensive comparisons in current practice are made by neural judges. Language models are widely used to judge pairs of outputs (Zheng et al., 2023); here a judge reads two outputs and scores how different they are, as natural-language-inference and equivalence judges do in diversity benchmarks (Stasaski and Hearst, 2022; Zhang et al., 2025). Such a judge has no feature map, so the all-pairs average costs at least one judge call per pair. Its scores also need not satisfy the triangle inequality, so Theorem 7 need not apply (although the Claude table at $T = 1$ below violates none of its 124,992 triangle inequalities). Theorem 5, Corollary 6, Propositions 10 and 15, and the Horvitz–Thompson variance of Remark 19 do apply, because they hold for every table, and every judge table below is symmetric by construction. We sampled populations of $n = 300$ responses from Qwen2.5-0.5B-Instruct (Yang et al., 2024) and Llama-3.2-1B-Instruct (Grattafiori et al., 2024). Each of 18 populations answers one of three open-ended prompts (a one-sentence story about a robot, a name for a new coffee shop, and one sentence of advice for a first job) at temperature $T = 0.5$, 1, or 1.5, and three further populations mix prompts, mix generators, or plant 2% degenerate samples. Three more contain 300 responses each from Claude Opus 5.5 to the same prompts, with default sampling (the API no longer accepts a sampling temperature). A cross-encoder trained for semantic textual similarity (STS; Reimers and Gurevych, 2019; Cer et al., 2017) scored every pair in both orders, giving

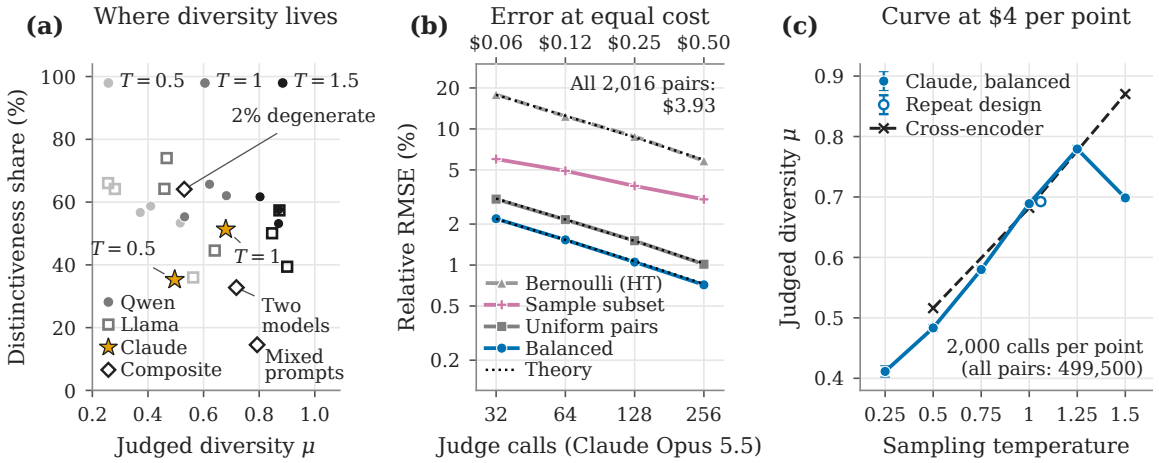


Figure 5: **Generated text scored by neural judges.** (a) Judged diversity μ and the share of the pairwise variance carried by distinctiveness. There are 21 cross-encoder tables with $n = 300$: Qwen (filled circles) and Llama (open squares), shaded by sampling temperature, and three composite populations (diamonds). The stars are the first 64 Qwen story responses at $T = 0.5$ and $T = 1$, judged by Claude Opus 5.5 instead of the cross-encoder. (b) Relative RMSE against the number of Claude calls for the 64 Qwen story responses at $T = 1$. Colored lines and markers are measured over 4,000 draws. Black dotted lines are the closed-form standard deviations of Theorem 5 and Corollary 6, and the square root of the Horvitz–Thompson variance of Remark 19; the closed form of Theorem 13 for the sample subset is not drawn. The top axis gives the API cost. (c) A diversity curve at linear cost: Claude-judged diversity of 1,000 Qwen2.5-0.5B-Instruct story responses per temperature, each point from one randomized round-robin design with $k = 4$ rounds (2,000 calls, about $4; all pairs would take 499,500), with the design-based 95% interval of Proposition 15 (t with 3 degrees of freedom), conditional on the 1,000 sampled responses; except at $T = 0.25$ the intervals are narrower than the markers. The open marker (offset for legibility) is an independent second design on the same responses. Dashed black: the exact all-pairs cross-encoder diversity of separate 300-response samples from the same generator and prompt (another scale and sample: compare shapes).

$d_{ij} = 1 - \bar{s}_{ij}$ with $\bar{s}_{ij} \in [0, 1]$ the mean of the two scores. That is 89,700 judge calls and 48–98 GPU-seconds per population. For the first 64 Qwen story responses at $T = 0.5$ and at $T = 1$, Claude Opus 5.5 (Anthropic) also rated all 2,016 pairs from 0 to 10, giving d_{ij} as the rating divided by 10 (Section S2.7).

Where judged diversity lives. Judged diversity rises with temperature: for Qwen, μ ranges over 0.37–0.52 at $T = 0.5$ and 0.80–0.87 at $T = 1.5$. Its anatomy, however, shows no consistent trend with temperature. For a single prompt and a small generator scored by the cross-encoder, distinctiveness carries 36–74% of the pairwise variance (Figure 5(a)),

because individual samples differ in how far they sit from all others, and the anatomy names them. At $T = 0.5$, the three least distinctive of the 300 samples under the cross-encoder are variants of one template ("A sleek humanoid robot named Echo…"), and two of the three least distinctive of the 64 under Claude are Echo variants as well. At $T = 1$, the two most distinctive samples under Claude are a response that switched to Chinese and an off-topic hacker story. Under the cross-encoder, on the same 64 responses, the hacker story ranks first and the Chinese response third; on all 300, a "Lorem ipsum" passage also enters the top three. Planting six degenerate samples among 300 raises the distinctiveness share from 53% to 64%. Mixing the three prompts at $T = 1$ lowers it from 55–66% to 15%, because the prompts act as roles and the structure becomes relational.

Frontier models as generators and as judges. Responses from a frontier model are far less diverse under this judge, as other measures also find (Zhang et al., 2025; Jiang et al., 2025). (The contrast is descriptive: the open models were capped at 40 new tokens, so 66–100% of their responses end mid-sentence, whereas the API model's are complete.) For the Claude Opus 5.5 populations, μ is only 0.22–0.41, below every Qwen population at $T = 1$ (0.53–0.68). The anatomy separates coffee-shop names that fall into a few modes (17% distinctiveness, so mostly relational) from advice that repeats one template with idiosyncratic exceptions (69%). The closed forms hold here as well; for example, the story prompt at $m = 2n$ gives 0.89% error against 0.89% predicted, and single-design intervals cover μ in 94.6–95.9% of draws. Claude and the cross-encoder score on different scales (a rating divided by 10 against one minus an STS probability), so we compare them by rank: they correlate 0.54–0.68 pair by pair, and their distinctiveness rankings of the 64 responses have Spearman correlations of 0.53 ($T = 1$) and 0.80 ($T = 0.5$). A third judge from another vendor, GPT-5 (OpenAI), scored the same 2,016 pairs at $T = 1$. It gives $\mu = 0.682$, correlates 0.64 with Claude pair by pair (Claude's own order-swapped retest gives 0.74), and ranks the 64 responses by distinctiveness with a Spearman correlation of 0.86 against Claude, with the same two responses on top. The closed forms hold on its table too (0.89% error at $m = 2n$ against 0.90% predicted).

Fewer judge calls, with error bars. Here the estimand is the diversity of the sampled responses, and errors refer to the design; claims about a generator add the sampling variance $4\zeta_1/n + O(n^{-2})$ (Section 7). Claude's order-swapped retest puts call noise at up to 46% of σ^2_{int} ($T = 1$), and 97% of such noise is relational, so $\kappa = 1.05$ is attenuated (about 1.9 without it). Because distinctiveness carries much of the variance ($\kappa = 0.56$–2.85), Corollary 6 predicts that uniform pair sampling needs 1.6–3.8 times as many judge calls as a balanced design on the Qwen and Llama single-prompt populations, and the measurements agree (4,000 design draws per cell). With $m = 2n$ pairs, the relative error on the Qwen story population at $T = 1$ is 0.49% for the balanced design (predicted 0.48%) and 0.77% for uniform pairs (predicted 0.78%). It is 4.1% for Bernoulli sampling with the Horvitz–Thompson estimator, and 3.4% for all pairs among a random subset of samples, the most common shortcut. With Claude as the judge on the $T = 1$ population (Figure 5(b)), 128 calls cost about \$0.25, against \$3.93 for all pairs, and give 1.05% error (predicted 1.07%); uniform pairs give 1.51% (predicted 1.50%). On the less diverse $T = 0.5$ population, the same 128 calls give 1.87% (predicted 1.90%) against 2.33% for uniform pairs. Over all 276 combinations of the 23 tables of Figure 5(a) (21 cross-encoder and 2 Claude-judged), the

three designs with a closed form, and the budgets $m \in \{n/2, n, 2n, 4n\}$ (4,000 draws each), the ratio of measured to predicted variance has median 0.999, and 90% of the ratios lie in $[0.96, 1.04]$. A single round-robin design with $k \in \{4, 8\}$ gives a 95% interval (Section 5.7) that covers μ in 94.5–95.7% of draws on every table of Figure 5. On the single-prompt populations of Figure 5(a), the same $m = 2n$ pairs rank samples by distinctiveness with a median Spearman correlation of 0.61–0.82 against the all-pairs ranking (0.49 on the three-prompt mixture).

A curve that all pairs cannot afford. Linear cost also makes analyses possible that the all-pairs average prices out. For 1,000 story responses at each of six temperatures, all pairs would take 499,500 Claude calls per temperature, about $1,000. One round-robin design with $k = 4$ rounds takes 2,000 calls (about $4), and through Proposition 15 it comes with its own error bar. Figure 5(c) shows the result. Judged diversity rises from 0.411 at $T = 0.25$ to 0.779 at $T = 1.25$, with 95% interval half-widths of 0.4–2.3% of $\widehat{\mu}$. It then falls to 0.698 at $T = 1.5$, where most responses degenerate into mixed-language noise and Claude rates pairs of them as less different from each other. The cross-encoder instead rates such noise as the most diverse ($\mu = 0.870$ at $T = 1.5$). Which notion of diversity is wanted is a modeling choice. The point here is that the cheap estimate is accurate enough to reveal the difference, for about $29 in total, including a repeat design at $T = 1$, instead of about $6,000 for all pairs at the six temperatures. The same calls also flag failures: at each temperature from 0.5 to 1, the response with the largest estimated distinctiveness $\widehat{a}_i$ (Proposition 10) is a refusal such as "I'm sorry, but I can't assist with that." The independent repeat design at $T = 1$ gave 0.692 (interval $[0.689, 0.696]$, against $[0.686, 0.691]$ for the first). The difference, 0.0035, is 2.5 standard errors when each design's own variance estimate is used; with $k = 4$ rounds these estimates have only 3 degrees of freedom (coefficient of variation about 40%), so the intervals are approximate, and more rounds give steadier error bars.

Sample sets at scale and the Vendi score. On 60,000 Fashion-MNIST images (Xiao et al., 2017), compared by the Euclidean distance between features of a small trained classifier, $m = n/2 = 30{,}000$ comparisons (0.0017% of the 1.8×10^9 pairs; 1,000 draws) give 0.169% error (predicted 0.167%). Uniform pairs give 0.207%, Bernoulli sampling 0.59%, and a subset of samples 1.96%. The order-2 Vendi score (Friedman and Dieng, 2023; Pasarkar and Dieng, 2024), which equals the Rényi-kernel-entropy mode count of Jalali et al. (2023), is $\mathrm{VS}_2 = n/\big(1 + (n-1)\,\bar{q}\big)$, where $\bar{q}$ is the average of $q_{ij} = s_{ij}^2$ for a similarity kernel with $s_{ii} = 1$. It is therefore a pairwise average in disguise, and the same comparisons estimate $\bar{q}$ without bias (the plug-in VS_2, convex in $\bar{q}$, is biased slightly upward). With a Gaussian kernel, the error is 0.29% for the balanced design, 0.34% for uniform pairs, and 2.9% for a subset. On text, with the judge's similarity as the kernel (not necessarily positive semidefinite, so VS_2 is then defined by the formula above), the error at $m = 2n$ is 2.1% for the balanced design against 3.0% for uniform pairs (cross-encoder, Qwen story population at $T = 1$), and 4.1% against 5.6% (Claude, $T = 1$). With the cross-encoder's similarity as the kernel, VS_2 is 1.6–2.7 for each set of 300 frontier-model responses, against 4.2–8.3 for Qwen at $T = 1$. Because this kernel gives substantial similarity to any two on-topic texts (a constant similarity s gives $\mathrm{VS}_2 \approx 1/s^2$ whatever the content), these values rank populations under one kernel rather than count distinct responses. For kernels with an explicit low-dimensional feature map, exact computation is already inexpensive. The gain is for

implicit kernels and judges: at $n = 10{,}000$, all pairs would take 5×10^7 judge calls, and a balanced design with $m = 2n$ takes 2×10^4.

5.7 Error bars and the anatomy from the same comparisons

The round-robin design (Section 4.2) also provides an error bar and the anatomy without extra comparisons. For even n, a round-robin schedule applied after a uniform relabeling partitions all N pairs into $n - 1$ rounds of $n/2$ disjoint pairs, so the round means average exactly to μ. Choosing the k rounds at random makes their means a simple random sample from these $n - 1$ values, and a t-interval over the k round means with the finite-population correction $1 - k/(n - 1)$ is the classical random-group interval (Wolter, 2007): its variance estimate is exactly unbiased (Proposition 15), and its coverage rests on approximate normality of the round means (Hájek, 1960). Over 26 tables (six synthetic families, the model population, and 19 VMAS tables: four trained teams and 14 reset variants from seven of the first nine training runs, plus one untrained $n = 100$ team; 2,000 draws per table), this interval covers μ in 95.0–95.2% of draws at the nominal 95% level for $k \in \{2, 4, 8, 16\}$, and every table lies between 93.6% and 96.4% (at $k = 16$, only the 13 tables with $n \geq 32$ admit the design). The usual Wald interval for uniform pairs undercovers on populations with an outlier (88.3% at $k = 2$). At $k = 4$ the balanced interval on the one-outlier table is 8× narrower than the Wald interval (0.72% of μ against 5.93%). With $k = 2$ the t-quantile on one degree of freedom makes intervals wide, and on the three $n = 64$ judge tables of Section 5.6 they also undercover slightly (91.9–94.1%), so we recommend $k \geq 4$ when error bars are needed. The same comparisons also rank individuals by distinctiveness (Proposition 10). On these 19 VMAS tables, the Spearman correlation between $\widehat{a}$ and a (over the first 300 draws per table) is 0.74 with four comparisons per agent and 0.91 with sixteen (on the six tables with $n \geq 32$; Figure S2 in Section S3).

5.8 Closed-loop diversity control

Proposition 11 predicts that a diversity controller driven by a sparse estimate tracks its target with relative error $-e/(1 + e)$, where e is the estimator's relative error, so to first order the tracking error is the estimation error with its sign reversed. We test this in two ways: a controlled simulation that isolates the mechanism, and DiCo training runs in VMAS.

Simulated control with drifting policies (no learning). A DiCo-style controller rescales the agent-specific action components of n agents by the factor $c_t = \mu_{\text{des}}/\widehat{\mu}_t$ at every step, while the components follow a random drift between steps. The distance is the mean Euclidean distance between agent-specific action means over a batch of 32 observations, which is W_2 for Gaussians with a shared covariance and is homogeneous of degree one, so the realized diversity equals $\mu_{\text{des}}\, \mu_t/\widehat{\mu}_t$ identically. The simulation therefore shows how each design's estimation error becomes tracking error as the population drifts; it does not model learning. We run 300 steps for teams with equal agent scales (homogeneous teams, $\kappa = 0.10$–0.24) and heterogeneous teams, whose agents differ in scale ($\kappa = 8$–31), at $n = 50$ with $m = 125$ comparisons per step (round-robin, $k = 5$) and at $n = 500$ with $m = 2{,}000$ of 124,750 pairs ($k = 8$), on 10 independently drawn populations per setting (Section S2.4).

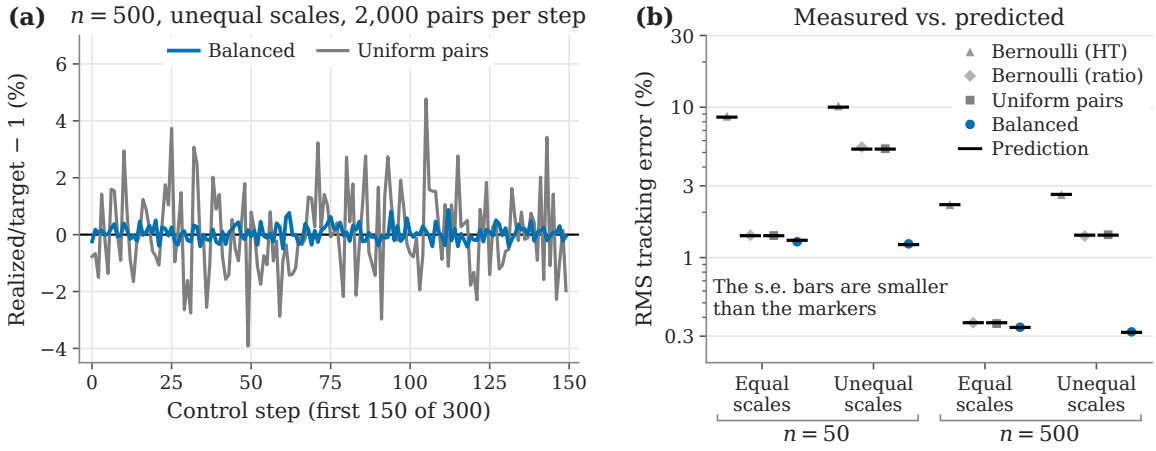


Figure 6: **A diversity controller inherits the estimator's error, and balanced designs shrink it.** (a) Realized diversity relative to the target for a DiCo-style controller on a 500-agent team whose agents differ in scale, using 2,000 comparisons per step (1.6% of all pairs); first 150 of 300 steps of one population. (b) Root-mean-square tracking error over 300 steps for teams of $n = 50$ or 500 agents with equal or unequal scales, mean ± standard error over 10 populations per setting. Short black bars through the markers are first-order predictions (the RMS of the relative estimation error e) from Theorem 5 (balanced), Corollary 6 (uniform pairs; for the Bernoulli ratio estimator, the uniform-pairs value at $m = pN$, a first-order lower bound), and Remark 19 (Horvitz–Thompson); Proposition 11 maps e to the tracking error $-e/(1+e)$. In this simulation, all-pairs control has zero error by construction.

Figure 6 shows that every measured tracking error matches its first-order prediction. For heterogeneous teams at $n = 500$, the root-mean-square tracking error is $0.320 \pm 0.004\%$ with the balanced design (predicted 0.318%), $1.42 \pm 0.05\%$ with uniform pairs (predicted 1.41%), $1.39 \pm 0.04\%$ with Bernoulli sampling and the ratio estimator (first-order lower bound 1.41%, met within Monte Carlo error), and $2.63 \pm 0.05\%$ with the Horvitz–Thompson estimator (predicted 2.63%). For heterogeneous teams at $n = 50$ the balanced design cuts the error from 5.3% (uniform pairs) to 1.23%, a factor of 4.3. With homogeneous scales, balanced and uniform designs perform alike (1.27% versus 1.40% at $n = 50$, and 0.344% versus 0.364% at $n = 500$), as Corollary 6 predicts. Where the estimation error is large, the mean tracking error is positive and close to the mean squared estimation error, which is the second-order bias of Proposition 11. For heterogeneous teams at $n = 50$ it is $+0.98 \pm 0.23\%$ with Horvitz–Thompson weighting (squared RMSE 1.05%), $+0.33 \pm 0.14\%$ with uniform pairs (0.28%), and $+0.02 \pm 0.02\%$ with the balanced design (0.015%).

DiCo in multi-agent reinforcement learning. We next replace the all-pairs SND inside DiCo (Bettini et al., 2024b) with a sparse estimate and train teams of $n = 50$ agents with independent PPO on VMAS Dispersion for 167 iterations, at the set points

$\mu_{\text{des}} \in \{0.12, 0.14, 0.15\}$ with three seeds each. Every cell is trained with all-pairs SND, with Bernoulli sampling at $p = 0.1$ and the ratio estimator (122.5 of 1,225 pairs per call in expectation), and, at the two lower set points, with the randomized balanced design: a fresh random 5-regular comparison graph at every call (125 pairs; Theorem 5). By Jensen's inequality and Corollary 6, Bernoulli sampling is per call at least $0.96\,(1 + \kappa)$ times as variable as the balanced design. The Bernoulli and all-pairs runs of a cell share seeds and environment initializations, so their rollouts diverge only once the estimates differ; the balanced runs reuse the seeds on a later software stack with evaluation rollouts disabled, on which two all-pairs runs reproduce the earlier ones to within seed-to-seed variation (Section S2.5), so comparisons with them match cells by seed and set point only and include any stack effect. The DiCo distance is homogeneous of degree one, so Proposition 11 applies (Section S2.5).

Table 3 reports the final 50 iterations. An all-pairs audit of the controlled policies, never fed back to the controller, finds their seed-averaged realized SND within 0.51% of both targets with the balanced design (at most 0.57% for a single seed), against 0.61% (0.71%) with Bernoulli sampling and 0.48% (0.54%) with all-pairs control at the same set points. We detect no change in reward, but three seeds give little power. The set points of one seed share initial conditions, and their reward differences are strongly correlated, so we treat seeds as the unit: the mean reward per step differs from all-pairs control by $(4.6 \pm 12.5) \times 10^{-3}$ with Bernoulli sampling and by $(9.1 \pm 11.4) \times 10^{-3}$ with the balanced design (mean $\pm$ standard error over three seeds; $t = 0.37$ and 0.80 on two degrees of freedom), and the 95% intervals exclude only differences beyond about -15% and $+18\%$ of the mean reward. The two sparse designs differ by $(1.1 \pm 4.1) \times 10^{-3}$ in reward (interval within $\pm 6\%$) and by -0.08 ± 0.10 percentage points in audit error (per-cell values in Table S8). These runs show that sparse estimation can replace all-pairs SND in DiCo without detectable loss, not that the choice of design matters in closed loop: with DiCo's soft update added to the simulation above (Table S7), the balanced design's $4.0\times$ smaller tracking error at $\tau = 1$ ($n = 50$, unequal scales) shrinks to $2.4\times$ at $\tau = 0.1$ and vanishes at $\tau \leq 0.03$, where the lag of the running average dominates every design, all pairs included. The metric cost per call falls 9.3–$9.6\times$ with Bernoulli sampling (19.7 to 2.1 ms on an RTX 4090) and 8.5–$9.0\times$ with the balanced design (196–205 to 23 ms on a shared Quadro RTX 5000), close to the $10\times$ and $9.8\times$ reductions in pairs. These times are for the reference DiCo implementation, which loops over pairs; a vectorized all-pairs computation takes about 0.6 ms at this n (Figure 7), so in MARL, sparse designs pay off with thousands of policies or with rollout-based comparisons.

5.9 Cost at scale

Figure 7(a) reports the wall-clock time of one complete diversity evaluation, including the forward passes of all n policies on a shared batch of 1,024 observations, on a single RTX 3080 GPU. All-pairs SND grows quadratically, from 0.6 ms at $n = 50$ to 3.1 s at $n = 5{,}000$. The balanced design grows linearly: at $n = 5{,}000$ and $k = 8$ it takes 24 ms, which is $131\times$ faster (below the $625\times$ ratio of pair counts, because both include the forward passes of all n policies), and 48 ms at $n = 10{,}000$. Figure 7(b) shows that accuracy improves at the same time. With k fixed, the relative standard deviation of the balanced estimate falls with n, to

Table 3: **DiCo with a sparse SND estimate: no detectable loss against all pairs.** ($n = 50$ agents, VMAS Dispersion, three seeds per set point, final 50 of 167 iterations). Balanced: a fresh random 5-regular comparison graph at every call (125 of 1,225 pairs); Bernoulli: $p = 0.1$ (122.5 pairs in expectation). Audit error: relative deviation of the realized all-pairs SND from the target, measured on 4,096 observations of each iteration's rollouts before they are used for training, and never fed back; all-pairs control also errs, because DiCo rescales with a running average ($\tau = 0.01$) of per-batch SND values. Reward: mean reward per step over the training batches, with its standard error over seeds. Balanced runs cover the two lower set points.

	Audit error (%)			Reward		
μ_{des}	balanced	Bernoulli	all pairs	balanced	Bernoulli	all pairs
0.12	0.44	0.50	0.48	0.307 ± 0.008	0.316 ± 0.008	0.306 ± 0.004
0.14	0.51	0.61	0.36	0.346 ± 0.002	0.335 ± 0.006	0.329 ± 0.012
0.15	–	0.43	0.46	–	0.339 ± 0.011	0.3412 ± 0.0004

0.054% at $n = 5{,}000$ and $k = 8$ (predicted 0.054%), because the relative variance is about $2\sigma_{\text{int}}^2/(\mu^2 nk)$ (Theorem 5) and at most $v_{n,k} < 2/(nk)$ for every metric (Theorem 7); the latter bound is loose here because independently initialized policies are nearly equidistant.

6 Related Work

Diversity as a mean pairwise dissimilarity. The average dissimilarity over all pairs of a population is one of the oldest and most widely used diversity statistics in science. Up to a factor $(n-1)/n$, it is Rao's quadratic entropy in ecology (Rao, 1982). Effective-number transformations of it and of related similarity-sensitive indices are standard there (Hill, 1973; Leinster and Cobbold, 2012; Chiu and Chao, 2014). In population genetics, nucleotide diversity is the mean number of pairwise sequence differences (Nei and Li, 1979; Tajima, 1983). Machine learning uses the same functional, or close relatives of it, throughout. It appears as disagreement-based ensemble diversity (Kuncheva and Whitaker, 2003), Self-BLEU for generated text (an average of multi-reference BLEU scores; Zhu et al., 2018), the average learned sentence similarity between pairs of responses (Tevet and Berant, 2021), and mean pairwise perceptual (LPIPS) distance for generated images (Zhu et al., 2017; Zhang et al., 2018), whose standard protocol already averages only 19 consecutive pairs of 20 samples, a sparse design of the kind analyzed here. It also appears as internal diversity of generated molecules (Polykovskiy et al., 2020), as the output diversity of aligned language models (Kirk et al., 2024), and as the diversity coefficient of training data (Miranda et al., 2025). For n equally weighted regression members, the ambiguity term of the squared-loss ensemble decomposition (Krogh and Vedelsby, 1995) is $(n-1)/(2n)$ times the mean pairwise squared distance between their predictions, and Wood et al. (2023) extend this diversity term to bias-variance-diversity decompositions of other losses. Kernel-based effective-number scores such as the Vendi score (Friedman and Dieng, 2023) are not pair averages in general. Their order-2 members (Pasarkar and Dieng, 2024), however, are monotone functions of a mean

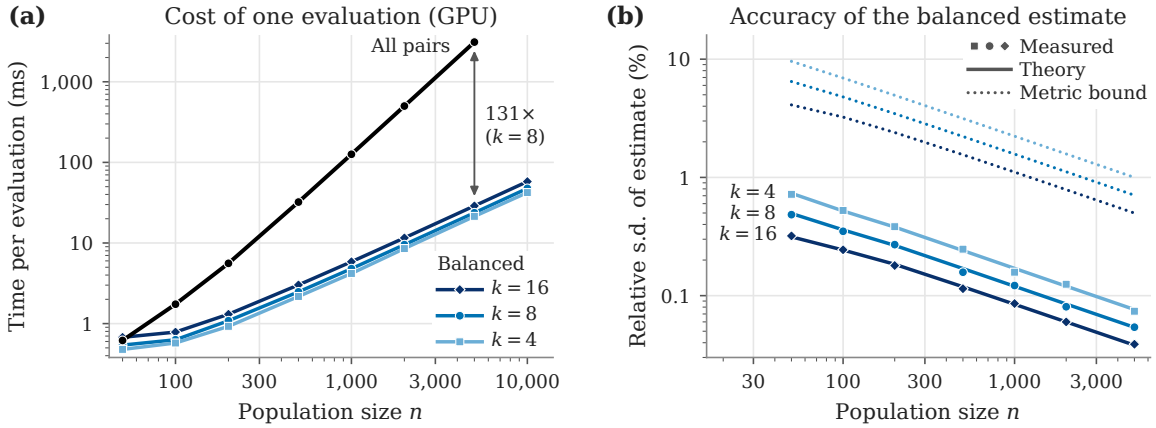


Figure 7: **Linear cost and improving accuracy at scale.** (a) Wall-clock time of one SND evaluation for n independently initialized Gaussian policies (median of repeated calls, RTX 3080). All pairs are measured up to $n = 5{,}000$; the arrow marks the 131× speedup of the balanced design with $k = 8$ at $n = 5{,}000$. (b) Relative standard deviation of the balanced estimate over 400 independent designs (markers: squares, circles, and diamonds for $k = 4$, 8, and 16; Monte Carlo standard errors are smaller than the markers), the closed-form prediction of Theorem 5 computed from the full table (solid lines), and the metric-universal bound $\sqrt{v_{n,k}}$ of Theorem 7 (dotted).

pairwise (squared) similarity, with the diagonal included, so the designs of Section 4 also cover them (Section 5.6). The order-2 Rényi kernel entropy of Jalali et al. (2023) is one such member, and Ospanov et al. (2024) approximate it and the Vendi score in linear time with random Fourier features, which requires a shift-invariant kernel on explicit inputs. In MARL, the same average is SND (Bettini et al., 2025). Except for Self-BLEU and Vendi scores of order other than 2, each of these statistics is a complete degree-two U-statistic over the population or a monotone function of one. When each comparison is expensive, as with rollouts, Wasserstein distances over many states, or judge-based comparisons (Zheng et al., 2023), its $\binom{n}{2}$ cost becomes the bottleneck that this paper removes.

Behavioral diversity and heterogeneity in MARL. Diversity in MARL has been quantified through policy and joint-action distributions (Yu et al., 2021; McKee et al., 2022; Hu et al., 2022), occupancy measures (Liu et al., 2021), trajectory distributions (Lupu et al., 2021), population determinants (Parker-Holder et al., 2020), and robotic heterogeneity indices (Balch, 2000; Twu et al., 2014; Prorok et al., 2017). A complementary line of work learns heterogeneous behavior through selective or masked parameter sharing (Christianos et al., 2021; Li et al., 2024), diversity regularization (Li et al., 2021), roles (Wang et al., 2020, 2021), heterogeneous graph policies (Bettini et al., 2023), and hypernetworks (Tessera et al., 2025); pairwise policy distances serve as diagnostics (Hu et al., 2024), and per-agent parameters are now standard (Zhong et al., 2024). SND is widely used in this line: DiCo holds it at a set point (Bettini et al., 2024b), it is used to study when heterogeneity pays

off (Bettini et al., 2024c; Amir et al., 2026), and DiCo-style control has group-aware and extremum-seeking variants (Yang et al., 2025; Patel and Jerath, 2026). Identity-free alternatives address agents that swap roles (Büchi et al., 2026), and diversity is also studied in societies of language-model agents (Huot et al., 2026). All of these pairwise measures evaluate every pair of agents. Our designs evaluate linearly many pairs with exact guarantees, and the anatomy adds a per-agent and relational reading of the same quantity (Section 5.4).

Incomplete U-statistics and design of comparisons. Averages of a degree-two kernel over a subset of pairs are incomplete U-statistics (Blom, 1976). Classical work mostly treats the data as i.i.d. draws and studies the variance about the population parameter. Blom (1976) and Lee (1982) give the exact variance of an incomplete U-statistic in terms of how many selected pairs share an observation, so for a degree-two kernel a design enters only through its degree sequence. For a non-degenerate kernel, designs in which every observation appears equally often recover the efficiency of the complete statistic once the number of pairs grows faster than n (Brown and Kildea, 1978; Lee, 1982, 1990), and designs built from orthogonal arrays can be asymptotically efficient even when it grows more slowly than n (Kong and Zheng, 2021). Recently, Miglioli and Awan (2025) gave efficient algorithms for constructing equireplicate designs and proved a Berry–Esseen bound for incomplete U-statistics of deterministic designs that also covers degenerate kernels. Random designs were analyzed by Janson (1984) and, conditionally on the data and in high dimensions, by Chen and Kato (2019). Incomplete U-statistics scale empirical risk minimization through randomly sampled tuples (Clémençon et al., 2016) and kernel two-sample testing through fixed designs, from the linear-time maximum mean discrepancy (MMD) over disjoint pairs (Gretton et al., 2012) to aggregated tests whose cost interpolates between linear and quadratic time (Schrab et al., 2022). Balanced sampling in survey statistics selects random samples, with prescribed inclusion probabilities, whose Horvitz–Thompson estimates of the totals of auxiliary variables equal the known population totals (Deville and Tillé, 2004). In that terminology, our balanced designs are balanced samples of pairs with respect to the n individual-incidence variables, and Theorem 5 gives their exact variance, which for general balanced samples is only approximated (Deville and Tillé, 2005). Horvitz–Thompson weighting (Horvitz and Thompson, 1952) underlies the Bernoulli designs used as baselines.

What changes for a fixed population. In our setting the population of n agents or models is fixed and, as in the finite-population theory of U-statistics (Nandi and Sen, 1963; Bloznelis and Götze, 2001), the target is the complete statistic itself rather than a superpopulation mean. Much of the classical machinery carries over (Remark S4). A regular design removes the main effects in every realization, just as, for i.i.d. data, it has the same Hájek projection as the complete statistic, and the variance of a relabeled design is a permutation moment of a quadratic-assignment (Mantel) statistic (Daniels, 1944; Mantel, 1967; Hubert and Schultz, 1976; Hubert, 1987), which Theorem 13 expresses through the anatomy. Two things change. First, the benchmark changes: classical efficiency is measured against the sampling error of the complete statistic about the population parameter, and against that benchmark fewer than n evaluations can suffice (Kong and Zheng, 2021), whereas estimating the complete statistic of a fixed population needs $\Omega(n)$ evaluations (Proposition 8, whose constructions follow Barhum et al., 2007). Second, the finite-population correction, which

disappears over an i.i.d. superpopulation (Remark S4), lets irregular designs such as landmarks do better when $\kappa < 2(n-1)/\big(n(n-3)\big)$. The normal limit of quadratic-assignment statistics (Barbour and Eagleson, 1986) justifies approximate intervals. When item parameters are estimated from pairwise comparisons, the topology of the comparison graph governs the minimax rate (Shah et al., 2016).

Sublinear estimation of average distance. In sublinear algorithms, Indyk (1999) showed that the average distance of n points in a metric space can be approximated within a factor $1 \pm \varepsilon$ using $O(n/\varepsilon^{7/2})$ distance queries. Barhum et al. (2007) showed that $O(n/\varepsilon^2)$ uniformly sampled pairs suffice, which is tight for uniform sampling. Chechik et al. (2015) reached $O(n + \varepsilon^{-2})$ queries with unbiased estimates built from a weighted sample of pairs, whose construction itself uses $O(n)$ distance computations. Chang (2017) analyzed a uniformly random perfect matching, our design with $k = 1$, for the metric 1-median problem; his variance bound needs a diameter condition (see also Chang, 2022) that he verifies for graph metrics, where it gives $(1 \pm \varepsilon)$ estimates of the average distance, while for general metrics he obtains one-sided constant-factor estimates. Theorem 7 with $k = 1$ makes a single random perfect matching a two-sided $(1 \pm \sqrt{8/n})$-approximation with probability at least $3/4$ for every pseudometric. Related average-parameter problems for graphs are treated by Feige (2006) and Goldreich and Ron (2008). Answering a question of Mendel and Naor (2015), Eskenazis et al. (2026) give a deterministic algorithm that queries $O(n)$ distances and approximates the average distance of a typical regular graph within a constant factor; they also show that no deterministic algorithm with $O(n)$ queries achieves a constant factor for all bounded-degree graphs. For several positive-definite kernels on Euclidean inputs, including the Gaussian kernel, the sum of all kernel entries, and hence a mean pairwise similarity, can be estimated in time sublinear in n by kernel density estimation (Backurs et al., 2021). That approach needs explicit inputs and positive definiteness. With only a dissimilarity oracle, $\Omega(n + \varepsilon^{-2})$ evaluations are necessary (Proposition 8). For constant failure probability, Theorem 7 attains this budget with a design that is non-adaptive and uniformly weighted and whose variance is known in closed form for every table. For failure probability η, the adaptive scheme needs only $O(n + \varepsilon^{-2} \log(1/\eta))$ evaluations (Chechik et al., 2015, Theorem 1.3), whereas the median of balanced designs pays the factor $\log(1/\eta)$ on the n term as well.

Decompositions of dyadic data. The anatomy is the two-way analysis of variance of a symmetric dyadic table, which in plant breeding is Griffing's diallel analysis of general and specific combining ability (Griffing, 1956). It coincides with the finite-population Hoeffding decomposition of a degree-two U-statistic (Nandi and Sen, 1963; Bloznelis and Götze, 2001), with the symmetric social relations model for interpersonal data (Kenny and La Voie, 1984), and with the spectral analysis of data on two-element subsets through the representation theory of the symmetric group (James, 1978; Diaconis, 1988). Related double-centering operations underlie classical multidimensional scaling (Torgerson, 1952; Gower, 1966) and distance covariance (Székely et al., 2007). The off-diagonal centering that defines R is exactly the U-centering behind unbiased distance covariance (Székely and Rizzo, 2014; Lyons, 2013), which Zhang (2026) recently characterized as a least-squares fit of additive endpoint effects, and Edelmann et al. (2020) bound the distance standard deviation by Gini's mean

difference, the population counterpart of Theorem 7. Crossing randomly numbered breeding lines in a circulant pattern, each with k others, is the circulant partial diallel cross of plant breeding (Kempthorne and Curnow, 1961). Bettini et al. (2025) already read per-agent contributions to SND off the row sums of the distance matrix; the distinctiveness a_i of Definition 1 is that row sum, centered and rescaled, and the resulting distinctiveness table $(a_i + a_j)$ is orthogonal to the level and to the relational part. For a given partition, the apportionment of quadratic entropy (Rao, 1982), the analysis of molecular variance (AMOVA; Excoffier et al., 1992), and permutational multivariate analysis of variance (PERMANOVA; Anderson, 2001) split a pair average into within-group and between-group terms. The anatomy needs no partition, and here it serves both as a diagnostic and as the currency in which the error of any comparison design is paid.

Expanders. The expander mixing lemma (Alon and Chung, 1988; Hoory et al., 2006) bounds bilinear forms of the adjacency matrix of a regular graph by its largest nontrivial eigenvalue in absolute value. Spectral sparsification (Spielman and Srivastava, 2011) preserves Laplacian quadratic forms with few reweighted edges. Random regular graphs are nearly Ramanujan (Friedman, 2008; Bordenave, 2020). Barhum et al. (2007) used the mixing lemma to give deterministic expander guarantees for ℓ_1-embeddable tables, which cover SND with diagonal Gaussian policies (Section 4.4). The same authors observe that the edge average of any regular graph is at most 2μ on metric tables, use the one-outlier metric to show that a design approximating every metric must be nearly regular (a metric precursor of Proposition 4), and show that no fixed design with $O(n)$ pairs approximates the average of every metric within a constant factor. Theorem 9, the mixing lemma applied to R through trace duality, holds for every table and charges only the relational residual. The fixed query graphs of Mendel and Naor (2015) and Eskenazis et al. (2026) give constant-factor guarantees for shortest-path metrics of typical random regular graphs. Once labels are randomized, expansion plays no role in the mean or the variance (Theorem 5).

7 Discussion

A recipe. The results combine into a short procedure for measuring the diversity of any population.

1. If exact evaluation is affordable, compute μ exactly. A squared Euclidean distance between explicit feature vectors, or an ℓ_1 distance, has an exact linear or near-linear-time formula (Proposition S1), and vectorized all-pairs SND takes about 0.6 ms for 50 agents on one GPU (Figure 7). Sparse designs pay off when each comparison is expensive (a rollout, a judge call, a transport distance) or when n reaches the thousands.

2. Otherwise, run Algorithm 1. If d is a metric, choose k from Theorem 7 for a guarantee that holds for every population: for a target relative standard deviation of order $n^{-1/2}$, k is a small constant, and $k = 8$ already gives, for every metric, a relative standard deviation of at most $\sqrt{v_{n,8}} < 0.5/\sqrt{n}$. For a sharper budget, or if d is not a metric (judge scores, for example), choose k from Theorem 5, with σ^2_{int} estimated from a first round-robin design as $\widehat{V}\, m(n-3)/(n-1-k)$, where $\widehat{V}$ is the unbiased variance estimate of Proposition 15, so that the estimate is unbiased by Theorem 5(iii).

3. Report the estimate together with the distinctiveness profile (Proposition 10) and an error bar, either from the round-robin design (even n, $k \geq 4$; Section 5.7), whose variance estimate is exactly unbiased (Proposition 15), or from several independent designs.
4. If a fixed graph is required, use a random k-regular expander (Theorem 9). If only interacting pairs matter, use the interaction graph, but report its value as *local* diversity and estimate its gap to μ with a small balanced design (Proposition 3).

Software. A NumPy package implements the anatomy, both balanced designs, the closed forms and the metric bound, median amplification, and distinctiveness estimates: `https://github.com/shawnray-research/Graph-SND`.

Fixed populations and sampled populations. Our guarantees are conditional on the population, which is the right target for a team, an ensemble, or a set of checkpoints. When the individuals are themselves a sample and the target is the expected dissimilarity of two independent draws, the complete average already has sampling variance $4\zeta_1/n + O(n^{-2})$, where $\zeta_1 = \mathrm{Var}_X\, \mathbb{E}[d(X, Y) \mid X]$ (Hoeffding, 1948). A balanced design adds $\sigma^2_{\mathrm{int}}(n-1-k)/(m(n-3)) \approx 2\sigma^2_{\mathrm{int}}/(nk)$. If $\zeta_1 > 0$, a constant k therefore loses only a constant factor of efficiency, which tends to one as k grows, in agreement with the classical analysis of balanced incomplete U-statistics (Blom, 1976; Brown and Kildea, 1978; Lee, 1990). For degenerate kernels ($\zeta_1 = 0$, for example points uniform on a circle with arc-length distance), the complete average has variance $O(n^{-2})$, and k must grow with n to keep the efficiency bounded away from zero.

Limitations. We treat each d_{ij} as exact. In SND, d_{ij} is itself an average over sampled observations. Its noise adds a term that no design removes, and that term is present in the all-pairs value as well. In our VMAS teams this noise is small: recomputing the tables of four final teams from three independent rollouts and observation pools (two Dispersion and two Navigation teams) changes μ by at most 2.3% and the distinctiveness share by at most 6 points, far less than the falls of 23–62 points in the distinctiveness share along the Dispersion trajectories. Judge scores are noisier: Claude's ratings of the same pairs in swapped order correlate at 0.74 (Section S2.7). Our closed forms treat the observed table as fixed; if each call carries independent noise of average variance $\bar{\tau}^2$, a design with m fresh calls adds about $\bar{\tau}^2/m$ to its variance, against $\bar{\tau}^2/N$ for all pairs. Judge scores also need not satisfy the triangle inequality, so Theorem 7 does not cover them in general (its bounded-range variant in Section 4.3 does); the results that hold for every table (Theorem 5, Corollary 6, and Propositions 10 and 15) still do. The fixed-design guarantee of Theorem 9 is a worst case over labelings and is conservative when the relational part has high rank. Our reinforcement learning experiments use VMAS, mostly with independent PPO, and even the seven longer runs (348–766 iterations) had not clearly converged, so the anatomy describes diversity during training rather than at convergence. The balanced-design DiCo runs cover two of the three set points. Finally, the anatomy describes how diversity is organized, not whether a given distinctiveness or relational share is desirable; that depends on the task.

8 Conclusion

Average pairwise dissimilarity is one of the most widely used measures of diversity in machine learning and the sciences. We used its canonical anatomy, consisting of a level, the distinctiveness of each individual, and relational structure, to show exactly what any sparse set of comparisons gets wrong. Comparing every individual equally often eliminates the distinctiveness error outright. Randomizing the labels then gives an unbiased estimate whose closed-form variance depends on relational structure alone, and a sharp inequality for metrics bounds that variance, so that $O(n + \varepsilon^{-2})$ comparisons fixed in advance give relative error ε with constant probability, the optimal rate, which to our knowledge was previously reached only adaptively. The same comparisons estimate every individual's distinctiveness, and a diversity controller inherits the estimator's accuracy to first order; in DiCo training of 50 agents, a balanced design that compares each agent with five others per call showed no detectable loss of tracking or reward relative to all-pairs control. On multi-agent teams, classifier populations, and judge-scored language-model outputs, balanced designs were the most reliable choice at equal cost, and the anatomy itself was informative: it separated two multi-agent tasks by how their diversity developed, recovered the training recipes of a classifier population, and flagged refusals among generated responses. In practice, once n exceeds ε^{-2}, measuring the diversity of a population costs a few comparisons per individual, whether a comparison is a Wasserstein distance between two policies, a Jensen–Shannon distance between two models, or (empirically, Section 5.6) a call to a language-model judge. Open questions include recovering role structure from the observed residuals, which is a matrix-completion problem on a random regular graph, and designs for comparisons that are noisy and must be re-queried.

Acknowledgments and Disclosure of Funding

The author thanks Matteo Bettini and the Prorok Lab for the open-source releases of SND, DiCo, and VMAS on which the multi-agent experiments build. An earlier version of this work was circulated as a preprint (Ray, 2026), which studied edge sampling and expander aggregation of SND. The present article supersedes it: the anatomy of diversity, the balanced-design theory with exact variances and optimal budgets, and the cross-domain experiments are new, and the forwarding-index analysis of the preprint is replaced by Theorem 9. This work received no external funding, and the author declares no competing interests.

Appendix A. Proofs

Throughout, $n \geq 4$, $N = \binom{n}{2}$, $\mu = \mu(D)$ and $S_i = \sum_{j \neq i} d_{ij}$. The vector a and matrix R are as in Definition 1. Sums over $i < j$ run over $\mathcal{P}$. Sums over $i \neq j$ run over ordered pairs, so $\sum_{i \neq j} f_{ij} = 2 \sum_{i<j} f_{ij}$ for symmetric f. For $f \in \mathbb{R}^{\mathcal{P}}$, f_e is its value on the pair e, and $\langle f, g \rangle = \sum_{i<j} f_{ij} g_{ij}$. We identify R with the vector $r = (r_{ij})_{i<j} \in \mathbb{R}^{\mathcal{P}}$.

A.1 Proof of Theorem 2

Proof *(i) Existence.* By construction, $d_{ij} = \mu + a_i + a_j + r_{ij}$ for $i \neq j$. Since $\sum_i S_i = 2N\mu = n(n-1)\mu$, we have $\sum_i a_i = \big(\sum_i S_i - n(n-1)\mu\big)/(n-2) = 0$. Using $\sum_{j\neq i} a_j = -a_i$,

$$\sum_{j\neq i} r_{ij} = S_i - (n-1)\mu - (n-1)a_i - \sum_{j\neq i} a_j = S_i - (n-1)\mu - (n-2)a_i = 0.$$

Uniqueness. Suppose $d_{ij} = \mu' + a'_i + a'_j + r'_{ij}$ with $\sum_i a'_i = 0$ and $\sum_{j\neq i} r'_{ij} = 0$ for every i. Summing over $i < j$ gives $N\mu = N\mu' + (n-1)\sum_i a'_i + \frac{1}{2}\sum_i \sum_{j\neq i} r'_{ij} = N\mu'$, so $\mu' = \mu$. Summing over $j \neq i$ gives $S_i = (n-1)\mu + (n-1)a'_i + \sum_{j\neq i} a'_j = (n-1)\mu + (n-2)a'_i$. Hence $a'_i = a_i$, and then $r' = r$.

(ii) Energy identity. Write $\tilde{a}_{ij} = a_i + a_j$. Then $\langle \mathbf{1}, \tilde{a}\rangle = (n-1)\sum_i a_i = 0$ and $\langle \mathbf{1}, r\rangle = \frac{1}{2}\sum_i \sum_{j\neq i} r_{ij} = 0$. Moreover,

$$\langle \tilde{a}, r\rangle = \tfrac{1}{2}\sum_{i\neq j}(a_i + a_j)r_{ij} = \sum_i a_i \sum_{j\neq i} r_{ij} = 0.$$

Hence $\sum_{i<j}(d_{ij} - \mu)^2 = \|\tilde{a}\|^2 + \|r\|^2$, that is, $N\sigma^2_{\text{tot}} = N\sigma^2_{\text{main}} + N\sigma^2_{\text{int}}$. Finally,

$$\begin{aligned}\sum_{i<j}(a_i + a_j)^2 &= (n-1)\sum_i a_i^2 + 2\sum_{i<j} a_i a_j \\ &= (n-1)\sum_i a_i^2 + \Big(\sum_i a_i\Big)^2 - \sum_i a_i^2 = (n-2)\sum_i a_i^2.\end{aligned}$$

(iii) Canonicity. The group $\mathfrak{S}_n$ acts on $\mathbb{R}^{\mathcal{P}}$ by $(\sigma f)_{ij} = f_{\sigma^{-1}(i)\sigma^{-1}(j)}$, which is the permutation module $M^{(n-2,2)}$ on 2-subsets. By Young's rule this module decomposes into irreducibles as $S^{(n)} \oplus S^{(n-1,1)} \oplus S^{(n-2,2)}$, each with multiplicity one, of dimensions 1, $n-1$, and $\binom{n}{2} - n = n(n-3)/2$ (James, 1978; Diaconis, 1988). The constants form the trivial summand $S^{(n)}$. The map $a \mapsto (a_i + a_j)_{i<j}$ is defined on $\{a \in \mathbb{R}^n : \sum_i a_i = 0\}$, which carries the standard representation $S^{(n-1,1)}$. It is $\mathfrak{S}_n$-equivariant. It is also injective for $n \geq 3$: if $a_i + a_j = 0$ for all $i \neq j$, then for distinct i, j, l we get $a_i = -a_j = a_l = -a_i$, so $a = 0$. In a multiplicity-free representation, every invariant subspace is a direct sum of isotypic components, because by Schur's lemma an irreducible invariant subspace projects to zero on every summand of a different isomorphism type. The image is an invariant copy of $S^{(n-1,1)}$, so it is the $S^{(n-1,1)}$ summand. The residual space $\{R : R\mathbf{1} = 0\}$ is invariant and, by (ii), orthogonal to the first two summands. It is the kernel of the n linear maps $R \mapsto \sum_{j\neq i} r_{ij}$ $(i \in [n])$, which are linearly independent by the injectivity argument above, so its dimension is $N - n = n(n-3)/2$. It is therefore exactly the orthogonal complement of the first two summands, namely the remaining summand $S^{(n-2,2)}$. The decomposition into irreducible invariant subspaces is unique, and it coincides with the anatomy. ■

A.2 Proof of Proposition 3

Proof Substituting Theorem 2(i) into $\widehat{\mu}_G$ gives

$$\widehat{\mu}_G = \frac{1}{W}\sum_{e=\{i,j\}\in E} w_e(\mu + a_i + a_j + r_e) = \mu + \frac{1}{W}\sum_i \delta_i a_i + \frac{1}{W}\sum_{e\in E} w_e r_e.$$

The middle term arises because individual i contributes a_i once per incident edge e, with weight w_e. Since $\sum_i a_i = 0$, we have $\sum_i \delta_i a_i = \sum_i (\delta_i - \bar{\delta}) a_i$. ■

A.3 Proof of Proposition 4

Proof For $D \in \mathcal{D}_\alpha$, Proposition 3 gives $\widehat{\mu}_G - \mu = W^{-1}\langle \delta - \bar{\delta}\mathbf{1}, a\rangle_{\mathbb{R}^n}$. Cauchy–Schwarz gives $|\widehat{\mu}_G - \mu| \le \alpha \|\delta - \bar{\delta}\mathbf{1}\|_2 / W$. If $\delta \equiv \bar{\delta}$, both sides vanish. Otherwise let $a^\star$ be $\delta - \bar{\delta}\mathbf{1}$ rescaled to Euclidean norm α. This vector sums to zero because $\sum_i \delta_i = 2W = n\bar{\delta}$. Fix $\mu^\star \ge 2\alpha$ and set $d_{ij} = \mu^\star + a_i^\star + a_j^\star$ for $i \ne j$. Then $d_{ij} \ge \mu^\star - 2\alpha \ge 0$. The triangle inequality $d_{ij} \le d_{il} + d_{lj}$ reduces to $0 \le \mu^\star + 2a_l^\star$, which holds since $|a_l^\star| \le \alpha$. By uniqueness in Theorem 2(i), this pseudometric table has level $\mu^\star$, distinctiveness $a^\star$ and $R = 0$, and it attains the bound. The error is independent of a for all tables if and only if $\delta - \bar{\delta}\mathbf{1} = 0$. ■

A.4 Proof of Theorem 5 and Corollary 6

Lemma 12 (Residual correlations) *Draw (e, f) uniformly from the ordered pairs of elements of $\mathcal{P}$ that share exactly one individual, and (e', f') uniformly from the ordered pairs of disjoint elements of $\mathcal{P}$. Then*

$$\mathbb{E}[r_e r_f] = -\frac{\sigma_{\text{int}}^2}{n-2}, \qquad \mathbb{E}[r_{e'} r_{f'}] = \frac{\sigma_{\text{int}}^2}{\binom{n-2}{2}}.$$

Proof *Shared individual.* Ordered pairs sharing exactly one individual correspond bijectively to ordered triples (i, j, l) of distinct individuals, with shared individual i, $e = \{i, j\}$ and $f = \{i, l\}$. There are $n(n-1)(n-2)$ of them. Since $R\mathbf{1} = 0$,

$$\sum_i \sum_{j \ne i} \sum_{l \notin \{i,j\}} r_{ij} r_{il} = \sum_i \Big[\Big(\sum_{j \ne i} r_{ij} \Big)^2 - \sum_{j \ne i} r_{ij}^2 \Big] = -\sum_{i \ne j} r_{ij}^2 = -2N\sigma_{\text{int}}^2.$$

So $\mathbb{E}[r_e r_f] = -2N\sigma_{\text{int}}^2 / (n(n-1)(n-2)) = -\sigma_{\text{int}}^2/(n-2)$.

Disjoint pairs. Fix $e = \{i, j\}$. Summing r_{kl} over all ordered (k, l) with $k \ne l$ gives 0. Let A be the set of ordered pairs with $k \in \{i, j\}$ and B the set with $l \in \{i, j\}$. Zero row and column sums give $\sum_A r_{kl} = \sum_B r_{kl} = 0$, and $A \cap B = \{(i, j), (j, i)\}$. By inclusion–exclusion, $\sum_{A \cup B} r_{kl} = -2r_{ij}$, so the ordered pairs disjoint from e sum to $2r_{ij}$. Each unordered pair appears twice among ordered pairs, so $\sum_{f \in \mathcal{P},\, f \cap e = \emptyset} r_f = r_e$. Averaging over the $N\binom{n-2}{2}$ ordered pairs of disjoint pairs gives $\mathbb{E}[r_{e'} r_{f'}] = \sum_e r_e^2 / \big(N\binom{n-2}{2}\big) = \sigma_{\text{int}}^2 / \binom{n-2}{2}$. ■

Proof of Theorem 5 *(i)* For every σ, $G = \sigma(H)$ is k-regular, so $\delta_i = \bar{\delta} = k$. The distinctiveness leakage term in Proposition 3 vanishes, which leaves $T - \mu = m^{-1} \sum_{e \in E(G)} r_e$.

(ii) For each fixed $e \in E(H)$, the pair $\sigma(e)$ is uniform on $\mathcal{P}$, because $\mathfrak{S}_n$ acts transitively on $\mathcal{P}$. Hence $\mathbb{E}[r_{\sigma(e)}] = N^{-1} \sum_{f \in \mathcal{P}} r_f = 0$ by Theorem 2(i), and $\mathbb{E}[T] = \mu$.

(iii) Let $X = \sum_{e \in E(H)} r_{\sigma(e)}$, so that $\mathrm{Var}(T) = \mathbb{E}[X^2]/m^2$ and

$$\mathbb{E}[X^2] = \sum_{e, f \in E(H)} \mathbb{E}\big[r_{\sigma(e)} r_{\sigma(f)}\big].$$

The group $\mathfrak{S}_n$ acts transitively on ordered pairs of pairs sharing exactly one individual, and also on ordered pairs of disjoint pairs. By the orbit–stabilizer theorem, a uniform σ

therefore maps any fixed such (e, f) to a uniform element of its orbit, and Lemma 12 applies. Because H is simple and k-regular, the double sum has three kinds of terms. There are m diagonal terms, each equal to σ^2_{int}. There are $nk(k-1) = 2m(k-1)$ ordered pairs of edges sharing one vertex, and $m(m-1) - 2m(k-1)$ ordered disjoint pairs. Therefore

$$\begin{aligned}\frac{\mathbb{E}[X^2]}{m\,\sigma^2_{\text{int}}} &= 1 - \frac{2(k-1)}{n-2} + \frac{2(m-2k+1)}{(n-2)(n-3)} \\ &= \frac{(n-2)(n-3) - 2(k-1)(n-3) + nk - 4k + 2}{(n-2)(n-3)} = \frac{n-1-k}{n-3},\end{aligned}$$

because $2m = nk$ and the numerator equals $(n-2)(n-1-k)$. Dividing by m^2 gives the claim. The resulting expression involves H only through (n, k), and it vanishes at $k = n-1$. ■

Proof of Corollary 6 The standard formulas for simple random sampling of m pairs without replacement (Cochran, 1977, Ch. 2) give

$$\operatorname{Var}(T_U) = \frac{s_d^2}{m}\Big(1 - \frac{m}{N}\Big) = \frac{\sigma^2_{\text{tot}}}{m} \cdot \frac{N-m}{N-1}, \qquad s_d^2 = \frac{1}{N-1}\sum_e (d_e - \mu)^2 = \frac{N}{N-1}\,\sigma^2_{\text{tot}}.$$

With $m = nk/2$, we have $N - m = n(n-1-k)/2$ and $N - 1 = (n-2)(n+1)/2$. Theorem 2(ii) gives $\sigma^2_{\text{tot}} = (1+\kappa)\sigma^2_{\text{int}}$. Dividing by the variance in Theorem 5(iii), which is positive for $k \le n-2$ when $\sigma^2_{\text{int}} > 0$, yields $(1+\kappa)\,n(n-3)/\big((n-2)(n+1)\big)$. ■

A.5 Randomized designs of arbitrary shape and round-robin error bars

Theorem 5 extends to uniformly relabeled designs of every shape. The extension gives an exact answer, valid for every population, to the question of Section 4: which pairs to compare.

Theorem 13 (Randomized designs of arbitrary shape) *Let H be any simple graph on $[n]$ with $m \ge 1$ edges and degrees $\delta_1, \dots, \delta_n$, and write $Q = \sum_u \delta_u^2$. Let σ be a uniformly random permutation and $T = m^{-1}\sum_{e \in E(\sigma(H))} d_e$. Then $\mathbb{E}[T] = \mu$ and*

$$\operatorname{Var}(T) = \frac{nQ - 4m^2}{2(n-2)\,m^2}\,\sigma^2_{\text{main}} + \frac{B(m,Q)}{m^2}\,\sigma^2_{\text{int}},$$
$$B(m,Q) = m - \frac{Q-2m}{n-2} + \frac{2(m^2+m-Q)}{(n-2)(n-3)}.$$

The variance depends on H only through (m, Q). Moreover $Q \ge 4m^2/n$, with equality exactly for regular H, which recovers Theorem 5. Because $\operatorname{Var}(T)$ is affine in Q, the following holds. If $\sigma^2_{\text{int}} > 0$, $m < N$, and a regular design with m edges exists ($n \mid 2m$), then among designs with m comparisons the regular ones minimize the variance if and only if $\kappa \ge 2(n-1)/\big(n(n-3)\big)$, the threshold of Corollary 6. Below this threshold the relational part dominates, and designs with more uneven degrees, such as stars or landmarks, have smaller variance.

Proof By Proposition 3 applied to $\sigma(H)$, $m(T-\mu) = X + Y$ with $X = \sum_u \delta_u a_{\sigma(u)}$ and $Y = \sum_{e\in E(H)} r_{\sigma(e)}$. Both have mean zero, because $a_{\sigma(u)}$ and $r_{\sigma(e)}$ are uniform over the individuals and the pairs, respectively.

Distinctiveness term. With $c_u = \delta_u - \bar{\delta}$ we have $\sum_u c_u = 0$ and, since $\sum_i a_i = 0$, $X = \sum_u c_u a_{\sigma(u)}$. For distinct u, v, $\mathbb{E}[a_{\sigma(u)}a_{\sigma(v)}] = \sum_{i\neq j} a_i a_j / \big(n(n-1)\big) = -\sum_i a_i^2 / \big(n(n-1)\big)$, and $\sum_{u\neq v} c_u c_v = -\sum_u c_u^2$. Substituting $\sum_u c_u^2 = Q - 4m^2/n$ and $\sum_i a_i^2 = N\sigma^2_{\text{main}}/(n-2)$ from Theorem 2(ii) then gives

$$\operatorname{Var}(X) = \Big(\frac{1}{n} + \frac{1}{n(n-1)}\Big) \sum_u c_u^2 \sum_i a_i^2 = \frac{1}{n-1} \sum_u c_u^2 \sum_i a_i^2 = \frac{nQ - 4m^2}{2(n-2)}\, \sigma^2_{\text{main}}.$$

Cross term. Let u be a vertex and $\{v, w\}$ an edge. If $u \in \{v, w\}$, then $\mathbb{E}[a_{\sigma(u)} r_{\sigma(v)\sigma(w)}]$ is proportional to $\sum_i a_i \sum_{j\neq i} r_{ij} = 0$. If $u \notin \{v, w\}$, the average runs over distinct i, j, l and is proportional to $\sum_i a_i \sum_{j\neq l,\, j,l\neq i} r_{jl} = \sum_i a_i \big(\sum_{j\neq l} r_{jl} - 2\sum_{l\neq i} r_{il} \big) = 0$. Hence $\mathbb{E}[XY] = 0$.

Relational term. As in the proof of Theorem 5, $\operatorname{Var}(Y)$ has three kinds of terms. There are m diagonal terms, each equal to σ^2_{int}. There are $\sum_u \delta_u(\delta_u - 1) = Q - 2m$ ordered pairs of edges sharing one vertex, each equal to $-\sigma^2_{\text{int}}/(n-2)$. There are $m(m-1) - (Q-2m)$ ordered disjoint pairs, each equal to $\sigma^2_{\text{int}} / \binom{n-2}{2}$ (Lemma 12). Summing gives $\operatorname{Var}(Y) = B(m, Q)\, \sigma^2_{\text{int}}$, and $\operatorname{Var}(T) = (\operatorname{Var}(X) + \operatorname{Var}(Y))/m^2$.

Shape. By Cauchy–Schwarz, $Q = \sum_u \delta_u^2 \geq (\sum_u \delta_u)^2/n = 4m^2/n$, with equality if and only if all degrees are equal. For a k-regular H, $Q = 2mk$, so $nQ = 4m^2$ and $B(m, Q)$ reduces to $m(n-1-k)/(n-3)$, which is Theorem 5(iii). The derivative of $m^2 \operatorname{Var}(T)$ with respect to Q is

$$\frac{n}{2(n-2)}\, \sigma^2_{\text{main}} - \frac{n-1}{(n-2)(n-3)}\, \sigma^2_{\text{int}},$$

which is non-negative if and only if $\kappa \geq 2(n-1)/\big(n(n-3)\big)$. ∎

Remark 14 (Why only two energies appear) *The form of Theorem 13 is a consequence of Theorem 2(iii). The second-moment operator $f \mapsto \mathbb{E}[\langle w_\sigma, f\rangle w_\sigma]$ of a design whose law is invariant under relabeling commutes with the action of $\mathfrak{S}_n$ on $\mathbb{R}^{\mathcal{P}}$. Here w_σ is the normalized indicator of $E(\sigma(H))$. By Schur's lemma, on a multiplicity-free representation such an operator acts as a scalar on each irreducible summand. The variance of every relabeling-invariant design is therefore a combination $c_1\sigma^2_{\text{main}} + c_2\sigma^2_{\text{int}}$ of the two energies, and the design enters only through the two scalars. Theorem 13 computes them.*

Proposition 15 (Unbiased variance estimate from one round-robin design) *Let n be even and $2 \leq k \leq n-2$. Relabel a fixed round-robin schedule (a partition of $\mathcal{P}$ into $n-1$ perfect matchings, called rounds) by a uniform permutation, select k of its $n-1$ rounds uniformly without replacement, let $\bar{d}_1, \ldots, \bar{d}_k$ be the mean dissimilarities of the selected rounds, and let $T = k^{-1}\sum_\ell \bar{d}_\ell$. Then the estimator*

$$\widehat{V} = \Big(1 - \frac{k}{n-1}\Big)\frac{s^2}{k}, \qquad s^2 = \frac{1}{k-1}\sum_{\ell=1}^{k} (\bar{d}_\ell - T)^2,$$

satisfies $\mathbb{E}[\widehat{V}] = \operatorname{Var}(T)$.

Proof Condition on the relabeling σ. The $n-1$ rounds partition $\mathcal{P}$ into perfect matchings of equal size $n/2$, so the average of their $n-1$ round means is exactly μ for every σ. Given σ, T is the mean of a simple random sample of k round means drawn without replacement from this finite population. Hence $\mathbb{E}[T \mid \sigma] = \mu$, and $\widehat{V}$ is the textbook unbiased estimator of $\mathrm{Var}(T \mid \sigma)$ (Cochran, 1977). Since $\mathbb{E}[T \mid \sigma]$ does not depend on σ, $\mathrm{Var}(T) = \mathbb{E}[\mathrm{Var}(T \mid \sigma)] = \mathbb{E}[\widehat{V}]$. ■

A.6 Proof of Theorem 7

Lemma 16 (Median amplification) *Let $T_1, \dots, T_L$ be independent estimates of μ. Suppose each one fails, meaning $|T_\ell - \mu| > \varepsilon\mu$, with probability at most $1/4$. Then*

$$\mathbb{P}\big(|\mathrm{med}(T_1, \dots, T_L) - \mu| > \varepsilon\mu\big) \le e^{-L/8},$$

which is at most η when $L \ge 8\ln(1/\eta)$.

Proof If the median lies outside $[(1-\varepsilon)\mu, (1+\varepsilon)\mu]$, then at least $L/2$ of the T_ℓ do. The number of such T_ℓ is a sum of independent indicators with mean at most $L/4$. By Hoeffding's inequality (Hoeffding, 1963), it reaches $L/2$ with probability at most $e^{-2L(1/4)^2} = e^{-L/8}$. ■

Proof of Theorem 7 *Step 1 (energy as a correlation).* By orthogonality (Theorem 2(ii)), $N\sigma_{\mathrm{int}}^2 = \sum_{i<j} r_{ij}^2 = \sum_{i<j} r_{ij}(\mu + a_i + a_j + r_{ij}) = \sum_{i<j} r_{ij} d_{ij}$.

Step 2 (averaged triangle inequality). For $i \ne j$ and each of the $n-2$ individuals $l \notin \{i, j\}$, $d_{ij} \le d_{il} + d_{lj}$. Summing over l gives

$$(n-2)\, d_{ij} \le (S_i - d_{ij}) + (S_j - d_{ij}), \qquad \text{so} \qquad d_{ij} \le \frac{S_i + S_j}{n}.$$

Step 3 (a residual bound). By (6), $a_i + a_j = \big(S_i + S_j - 2(n-1)\mu\big)/(n-2)$, so Step 2 gives

$$r_{ij} = d_{ij} - \mu - a_i - a_j = d_{ij} + \frac{n\mu - (S_i + S_j)}{n-2} \le d_{ij} + \frac{n\mu - n\, d_{ij}}{n-2} = \frac{n\mu - 2d_{ij}}{n-2}.$$

Step 4 (the energy bound). Since $d_{ij} \ge 0$, Steps 1 and 3 give

$$\begin{aligned} N\sigma_{\mathrm{int}}^2 = \sum_{i<j} r_{ij} d_{ij} &\le \sum_{i<j} \frac{(n\mu - 2d_{ij})\, d_{ij}}{n-2} = \frac{N}{n-2}\Big(n\mu^2 - 2\big(\mu^2 + \sigma_{\mathrm{tot}}^2\big)\Big) \\ &= N\Big(\mu^2 - \frac{2\sigma_{\mathrm{tot}}^2}{n-2}\Big), \end{aligned}$$

because $\sum_{i<j} d_{ij} = N\mu$ and $\sum_{i<j} d_{ij}^2 = N(\mu^2 + \sigma_{\mathrm{tot}}^2)$. This is the first bound. Since $\sigma_{\mathrm{int}}^2 \le \sigma_{\mathrm{tot}}^2$ by Theorem 2(ii), it gives $\sigma_{\mathrm{int}}^2\,\big(1 + 2/(n-2)\big) \le \mu^2$, that is, $\sigma_{\mathrm{int}}^2 \le (n-2)\mu^2/n$.

Tightness. Let n be even, split $[n]$ into two halves of size $n/2$, and set $d_{ij} = 1$ if i and j lie in different halves and $d_{ij} = 0$ otherwise. This cut metric is a pseudometric. Every row sum is $S_i = n/2$, so $a = 0$, and $\mu = (n/2)^2/N = n/(2(n-1))$. Since $d_{ij}^2 = d_{ij}$, we get $\sigma_{\mathrm{int}}^2 = \sigma_{\mathrm{tot}}^2 = \mu - \mu^2$, and

$$\mu(1-\mu) = \frac{n}{2(n-1)} \cdot \frac{n-2}{2(n-1)} = \frac{n(n-2)}{4(n-1)^2} = \frac{n-2}{n}\,\mu^2.$$

So the bound $\sigma^2_{\text{int}} \le (n-2)\mu^2/n$ is attained, and, because $\sigma^2_{\text{int}} = \sigma^2_{\text{tot}}$, so is the first bound of Step 4.

Consequences. If $\mu = 0$, then $D = 0$ and every claim is trivial, so assume $\mu > 0$. By Theorem 5(iii) with $m = nk/2$, and Step 4,

$$\frac{\mathrm{Var}(T)}{\mu^2} \le \frac{n-2}{n} \cdot \frac{n-1-k}{(n-3)m} = \frac{2(n-2)(n-1-k)}{n^2(n-3)k} = v_{n,k}.$$

Chebyshev's inequality gives the tail bound. Since $(n-2)(n-1-k) \le (n-2)^2 \le n(n-3)$ for $k \ge 1$ (the last inequality is equivalent to $n \ge 4$), we have $v_{n,k} \le 2/(nk)$. With $k \ge 8/(n\varepsilon^2)$ this gives $v_{n,k}/\varepsilon^2 \le 1/4$, and Lemma 16 applies. If the prescribed k reaches $n-1$, the design is complete and exact. For the budget, let $x = 8/(n\varepsilon^2)$. Then $\lceil \max\{1, x\} \rceil \le x+1$, rounding up to make nk even adds at most one more, and the cap at $n-1$ can only lower k. Hence every design has $k \le x+2$ and uses at most $nk/2 \le 4\varepsilon^{-2}+n$ evaluations. The median of $L \le 8\ln(1/\eta) + 1$ designs therefore uses $O\big((n+\varepsilon^{-2})(1+\log(1/\eta))\big)$ evaluations. ■

A.7 Proof of Proposition 8

The hard instances behind parts (i) and (ii) are standard in the sublinear-algorithms literature (Barhum et al., 2007; Goldreich and Ron, 2008); the proof, with explicit constants, is in Section S5.6 of the Supplementary Material.

A.8 Proof of Theorem 9

Proof *Part (i).* Let A be the adjacency matrix of G and $P = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$. Since R is symmetric with $R\mathbf{1} = 0$, we have $R = PRP$, and $\operatorname{tr} R = 0$ because $r_{ii} = 0$. Then $\sum_{e\in E} r_e = \frac{1}{2}\langle A, R\rangle_F$. For every $c \in \mathbb{R}$, $\langle A, R\rangle_F = \langle A - cI, PRP\rangle_F = \langle P(A-cI)P, R\rangle_F$. Because G is regular, $A\mathbf{1} = k\mathbf{1}$, so A preserves $\mathbf{1}^\perp$. The matrix $P(A-cI)P$ therefore has eigenvalues 0 on $\mathbf{1}$ and $\lambda_j - c$ ($j \ge 2$) on $\mathbf{1}^\perp$. With $c = (\lambda_2 + \lambda_n)/2$, its operator norm is $(\lambda_2 - \lambda_n)/2$. Trace duality (Hölder's inequality for Schatten norms) with $M = P(A-cI)P$, $|\langle M, R\rangle_F| \le \|M\|_{\text{op}}\|R\|_*$, gives $|\sum_{e\in E} r_e| \le (\lambda_2 - \lambda_n)\|R\|_*/4$. Dividing by $m = nk/2$ and applying Proposition 3, whose distinctiveness leakage vanishes for regular G, proves the first bound. For pseudometric tables, Theorem 7 gives $\|R\|_F^2 = \sum_{i\ne j} r_{ij}^2 = n(n-1)\sigma^2_{\text{int}} \le (n-1)(n-2)\mu^2$. Together with $\|R\|_* \le \sqrt{\operatorname{rank}(R)}\,\|R\|_F$, this yields $\|R\|_*/(n\mu) \le \sqrt{\operatorname{rank}(R)\,(n-1)(n-2)}/n$ when $\mu > 0$. The final claim follows from Friedman's theorem (Friedman, 2008; Bordenave, 2020): $\max(|\lambda_2|, |\lambda_n|) \le 2\sqrt{k-1} + o(1)$ with high probability.

Part (ii). Let Λ be diagonal and $\lambda = \max(|\lambda_2|, |\lambda_n|)$. Since $\operatorname{diag}(A) = 0$, we have $\langle A, \Lambda\rangle_F = 0$, so $\langle A, R\rangle_F = \langle A, R+\Lambda\rangle_F$. Regularity gives $A = \frac{k}{n}\mathbf{1}\mathbf{1}^\top + PAP$. Because $R\mathbf{1} = 0$ and P is an orthogonal projection,

$$\langle \mathbf{1}\mathbf{1}^\top, R+\Lambda\rangle_F = \mathbf{1}^\top R\mathbf{1} + \operatorname{tr}\Lambda = \operatorname{tr}\Lambda, \qquad \langle PAP, R+\Lambda\rangle_F = \langle PAP, P(R+\Lambda)P\rangle_F.$$

The matrix PAP has eigenvalues $0, \lambda_2, \dots, \lambda_n$, so its operator norm is λ. Hölder's inequality then gives

$$\Big|\sum_{e\in E} r_e\Big| \le \frac{1}{2}\Big(\frac{k}{n}|\operatorname{tr}\Lambda| + \lambda\,\|P(R+\Lambda)P\|_*\Big),$$

and dividing by $m = nk/2$, as in part (i), yields the bound of part (ii). ■

Remark 17 (Roles and the diagonal correction) *Suppose the population has K roles and the relational structure is constant within and between roles. Then $R = F - \mathrm{Diag}(F)$ for a block-constant matrix F with $\operatorname{rank} F \le K$. Because of its zero diagonal, R itself generically has rank $n-1$, so part (i) gives little. Choosing $\Lambda = \mathrm{Diag}(F)$ in part (ii) gives $P(R+\Lambda)P = PFP$, whose rank is at most K. Hence $\|PFP\|_* \le \sqrt{K}\,\|PFP\|_F$, while the trace term $|\operatorname{tr} F|/n^2 \le \max_i |F_{ii}|/n$ vanishes as n grows. If every role has at least two members, each F_{ii} equals an off-diagonal entry r_{ij} of row i (any $j \ne i$ in the role of i), so $\sum_i F_{ii}^2 \le \|R\|_F^2$ and*

$$\|PFP\|_F \le \|F\|_F \le \sqrt{2}\,\|R\|_F = \sqrt{2n(n-1)}\,\sigma_{\mathrm{int}}.$$

Part (ii) then gives

$$\Big|\frac{\widehat{\mu}_G}{\mu} - 1\Big| \le \frac{\lambda\sqrt{2K}\,\sigma_{\mathrm{int}}}{k\mu} + \frac{\max_i |F_{ii}|}{n\mu},$$

which is $O(\sqrt{K/k})\,\sigma_{\mathrm{int}}/\mu + O(1/n)$ when $\lambda \le 2\sqrt{k-1} + o(1)$ and $\max_i |F_{ii}| = O(\mu)$. Alternatively, the pseudometric bound of part (i) can be stated with the effective rank $r_{\mathrm{eff}}(R) = \|R\|_*^2/\|R\|_F^2 \le \operatorname{rank}(R)$ in place of $\operatorname{rank}(R)$, because $\|R\|_* = \sqrt{r_{\mathrm{eff}}(R)}\,\|R\|_F$ by definition.

Remark 18 (No fixed sparse design is uniformly accurate) *Randomization cannot be dropped from Theorem 7. Let G be a fixed connected k-regular graph and let D be its own shortest-path metric. Then every compared pair has $d_e = 1$, so $\widehat{\mu}_G = 1$. But at most $1 + k + \cdots + k^t$ individuals lie within distance t of any individual, so at least half of all pairs are at distance $\Omega(\log n/\log k)$, and $\mu = \Omega(\log n/\log k)$. The relative error of G therefore tends to -1 as $n \to \infty$ with k fixed. An adversary who knows the design can always choose such a table, whereas a uniformly relabeled design does not reveal its pairs in advance.*

A.9 Proofs of Propositions 10 and 11

Both proofs are short calculations, with simple random sampling for Proposition 10 and Chebyshev's and Markov's inequalities for Proposition 11; they are in Sections S5.7 and S5.8 of the Supplementary Material.

A.10 Variance of Bernoulli sampling

Remark 19 (Bernoulli sampling) *Under Bernoulli sampling, each pair e is included independently with probability p; its inclusion indicator is $\xi_e \sim \mathrm{Bernoulli}(p)$. The Horvitz–Thompson estimator $\widehat{\mu}_{\mathrm{HT}} = (pN)^{-1}\sum_{e\in\mathcal{P}} \xi_e d_e$ is unbiased. Its variance is exactly*

$$\mathrm{Var}(\widehat{\mu}_{\mathrm{HT}}) = \frac{p(1-p)}{p^2N^2}\sum_e d_e^2 = \frac{1-p}{pN}\big(\mu^2 + \sigma_{\mathrm{tot}}^2\big),$$

because $N^{-1}\sum_e d_e^2 = \mu^2 + \sigma_{\mathrm{tot}}^2$. Unlike every other design in Table 1, its variance contains the level μ^2 itself, so it is not even exact on a constant table.

References

N. Alon and F. R. K. Chung. Explicit construction of linear sized tolerant networks. *Discrete Mathematics*, 72(1–3):15–19, 1988.

M. Amir, M. Bettini, and A. Prorok. When is diversity rewarded in cooperative multi-agent learning? In *International Conference on Learning Representations (ICLR)*, 2026.

M. J. Anderson. A new method for non-parametric multivariate analysis of variance. *Austral Ecology*, 26(1):32–46, 2001.

A. Backurs, P. Indyk, C. Musco, and T. Wagner. Faster kernel matrix algebra via density estimation. In *International Conference on Machine Learning (ICML)*, pages 500–510, 2021.

R. A. Bailey. Strata for randomized experiments (with discussion). *Journal of the Royal Statistical Society, Series B*, 53(1):27–78, 1991.

R. A. Bailey. *Association Schemes: Designed Experiments, Algebra and Combinatorics*, volume 84 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2004.

T. Balch. Hierarchic social entropy: An information theoretic measure of robot group diversity. *Autonomous Robots*, 8(3):209–238, 2000.

A. D. Barbour and G. K. Eagleson. Random association of symmetric arrays. *Stochastic Analysis and Applications*, 4(3):239–281, 1986.

K. Barhum, O. Goldreich, and A. Shraibman. On approximating the average distance between points. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM)*, volume 4627 of *Lecture Notes in Computer Science*, pages 296–310. Springer, 2007.

M. Bettini, A. Shankar, and A. Prorok. Heterogeneous multi-robot reinforcement learning. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 1485–1494, 2023.

M. Bettini, R. Kortvelesy, J. Blumenkamp, and A. Prorok. VMAS: A vectorized multi-agent simulator for collective robot learning. In *International Symposium on Distributed Autonomous Robotic Systems (DARS)*, volume 28 of *Springer Proceedings in Advanced Robotics*, pages 42–56. Springer, 2024a.

M. Bettini, R. Kortvelesy, and A. Prorok. Controlling behavioral diversity in multi-agent reinforcement learning. In *International Conference on Machine Learning (ICML)*, pages 3611–3636, 2024b.

M. Bettini, R. Kortvelesy, and A. Prorok. The impact of behavioral diversity in multi-agent reinforcement learning. arXiv:2412.16244, 2024c.

M. Bettini, A. Shankar, and A. Prorok. System Neural Diversity: Measuring behavioral heterogeneity in multi-agent learning. *Journal of Machine Learning Research*, 26(163): 1–27, 2025.

G. Blom. Some properties of incomplete U-statistics. *Biometrika*, 63(3):573–580, 1976.

M. Bloznelis and F. Götze. Orthogonal decomposition of finite population statistics and its applications to distributional asymptotics. *The Annals of Statistics*, 29(3):899–917, 2001.

C. Bordenave. A new proof of Friedman's second eigenvalue theorem and its extension to random lifts. *Annales Scientifiques de l'École Normale Supérieure*, 53(6):1393–1439, 2020.

R. C. Bose and T. Shimamoto. Classification and analysis of partially balanced incomplete block designs with two associate classes. *Journal of the American Statistical Association*, 47(258):151–184, 1952.

B. M. Brown and D. G. Kildea. Reduced U-statistics and the Hodges–Lehmann estimator. *The Annals of Statistics*, 6(4):828–835, 1978.

H. Büchi, M. Flageat, E. Sebastián, and A. Prorok. Events as triggers for behavioral diversity in multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. To appear; arXiv:2605.12388.

E. J. Candès and B. Recht. Exact matrix completion via convex optimization. *Foundations of Computational Mathematics*, 9(6):717–772, 2009.

D. Cer, M. Diab, E. Agirre, I. Lopez-Gazpio, and L. Specia. SemEval-2017 Task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In *International Workshop on Semantic Evaluation (SemEval)*, pages 1–14, 2017.

C.-L. Chang. Metric random matchings with applications. arXiv:1703.08433, 2017.

C.-L. Chang. On random perfect matchings in metric spaces with not-too-large diameters. *Theory of Computing Systems*, 66:847–860, 2022.

S. Chechik, E. Cohen, and H. Kaplan. Average distance queries through weighted samples in graphs and metric spaces: High scalability with tight statistical guarantees. In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques (APPROX/RANDOM)*, volume 40 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 659–679, 2015.

X. Chen and K. Kato. Randomized incomplete U-statistics in high dimensions. *The Annals of Statistics*, 47(6):3127–3156, 2019.

C.-H. Chiu and A. Chao. Distance-based functional diversity measures and their decomposition: A framework based on Hill numbers. *PLoS ONE*, 9(7):e100014, 2014.

F. Christianos, G. Papoudakis, M. A. Rahman, and S. V. Albrecht. Scaling multi-agent reinforcement learning with selective parameter sharing. In *International Conference on Machine Learning (ICML)*, pages 1989–1998, 2021.

S. Clémençon, I. Colin, and A. Bellet. Scaling-up empirical risk minimization: Optimization of incomplete U-statistics. *Journal of Machine Learning Research*, 17(76):1–36, 2016.

W. G. Cochran. *Sampling Techniques*. John Wiley & Sons, New York, third edition, 1977.

H. E. Daniels. The relation between measures of correlation in the universe of sample permutations. *Biometrika*, 33(2):129–135, 1944.

J.-C. Deville and Y. Tillé. Efficient balanced sampling: The cube method. *Biometrika*, 91 (4):893–912, 2004.

J.-C. Deville and Y. Tillé. Variance approximation under balanced sampling. *Journal of Statistical Planning and Inference*, 128(2):569–591, 2005.

P. Diaconis. *Group Representations in Probability and Statistics*, volume 11 of *Lecture Notes–Monograph Series*. Institute of Mathematical Statistics, Hayward, CA, 1988.

D. Edelmann, D. Richards, and D. Vogel. The distance standard deviation. *The Annals of Statistics*, 48(6):3395–3416, 2020.

A. Eskenazis, M. Mendel, and A. Naor. An optimal algorithm for average distance in typical regular graphs. In *ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 742–757, 2026.

L. Excoffier, P. E. Smouse, and J. M. Quattro. Analysis of molecular variance inferred from metric distances among DNA haplotypes: Application to human mitochondrial DNA restriction data. *Genetics*, 131(2):479–491, 1992.

U. Feige. On sums of independent random variables with unbounded variance and estimating the average degree in a graph. *SIAM Journal on Computing*, 35(4):964–984, 2006.

D. Friedman and A. B. Dieng. The Vendi score: A diversity evaluation metric for machine learning. *Transactions on Machine Learning Research*, 2023.

J. Friedman. A proof of Alon's second eigenvalue conjecture and related problems. *Memoirs of the American Mathematical Society*, 195(910):1–100, 2008.

O. Goldreich and D. Ron. Approximating average parameters of graphs. *Random Structures & Algorithms*, 32(4):473–493, 2008.

J. C. Gower. Some distance properties of latent root and vector methods used in multivariate analysis. *Biometrika*, 53(3–4):325–338, 1966.

A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, et al. The Llama 3 herd of models. arXiv:2407.21783, 2024.

A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773, 2012.

B. Griffing. Concept of general and specific combining ability in relation to diallel crossing systems. *Australian Journal of Biological Sciences*, 9(4):463–493, 1956.

J. Hájek. Limiting distributions in simple random sampling from a finite population. *Publications of the Mathematical Institute of the Hungarian Academy of Sciences*, 5:361–374, 1960.

M. O. Hill. Diversity and evenness: A unifying notation and its consequences. *Ecology*, 54 (2):427–432, 1973.

W. Hoeffding. A class of statistics with asymptotically normal distribution. *The Annals of Mathematical Statistics*, 19(3):293–325, 1948.

W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

S. Hoory, N. Linial, and A. Wigderson. Expander graphs and their applications. *Bulletin of the American Mathematical Society*, 43(4):439–561, 2006.

D. G. Horvitz and D. J. Thompson. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685, 1952.

S. Hu, C. Xie, X. Liang, and X. Chang. Policy diagnosis via measuring role diversity in cooperative multi-agent RL. In *International Conference on Machine Learning (ICML)*, pages 9041–9071, 2022.

T. Hu, Z. Pu, X. Ai, T. Qiu, and J. Yi. Measuring policy distance for multi-agent reinforcement learning. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 834–842, 2024.

L. J. Hubert. *Assignment Methods in Combinatorial Data Analysis*, volume 73 of *Statistics: Textbooks and Monographs*. Marcel Dekker, New York, 1987.

L. J. Hubert and J. Schultz. Quadratic assignment as a general data analysis strategy. *British Journal of Mathematical and Statistical Psychology*, 29(2):190–241, 1976.

F. Huot, M. Kaisers, and M. Lapata. When is routing meaningful? Diversity and robustness in language model societies. arXiv:2607.09197, 2026.

P. Indyk. Sublinear time algorithms for metric space problems. In *ACM Symposium on Theory of Computing (STOC)*, pages 428–434, 1999.

M. Jalali, C. T. Li, and F. Farnia. An information-theoretic evaluation of generative models in learning multi-modal distributions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, pages 9931–9943, 2023.

G. D. James. *The Representation Theory of the Symmetric Groups*, volume 682 of *Lecture Notes in Mathematics*. Springer, Berlin, 1978.

S. Janson. The asymptotic distributions of incomplete U-statistics. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 66(4):495–505, 1984.

L. Jiang, Y. Chai, M. Li, M. Liu, R. Fok, N. Dziri, Y. Tsvetkov, M. Sap, A. Albalak, and Y. Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). arXiv:2510.22954, 2025.

O. Kempthorne and R. N. Curnow. The partial diallel cross. *Biometrics*, 17(2):229–250, 1961.

D. A. Kenny and L. La Voie. The social relations model. In L. Berkowitz, editor, *Advances in Experimental Social Psychology*, volume 18, pages 141–182. Academic Press, 1984.

R. Kirk, I. Mediratta, C. Nalmpantis, J. Luketina, E. Hambro, E. Grefenstette, and R. Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

X. Kong and W. Zheng. Design based incomplete U-statistics. *Statistica Sinica*, 31(3): 1593–1618, 2021.

A. Krogh and J. Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 7, pages 231–238, 1995.

L. I. Kuncheva and C. J. Whitaker. Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy. *Machine Learning*, 51(2):181–207, 2003.

B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 6402–6413, 2017.

A. J. Lee. On incomplete U-statistics having minimum variance. *Australian Journal of Statistics*, 24(3):275–282, 1982.

A. J. Lee. *U-Statistics: Theory and Practice*, volume 110 of *Statistics: Textbooks and Monographs*. Marcel Dekker, New York, 1990.

J. Lehman and K. O. Stanley. Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2):189–223, 2011.

T. Leinster and C. A. Cobbold. Measuring diversity: The importance of species similarity. *Ecology*, 93(3):477–489, 2012.

C. Li, T. Wang, C. Wu, Q. Zhao, J. Yang, and C. Zhang. Celebrating diversity in shared multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 3991–4002, 2021.

X. Li, L. Pan, and J. Zhang. Kaleidoscope: Learnable masks for heterogeneous multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 22081–22106, 2024.

X. Liu, H. Jia, Y. Wen, Y. Hu, Y. Chen, C. Fan, Z. Hu, and Y. Yang. Towards unifying behavioral and response diversity for open-ended learning in zero-sum games. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 941–952, 2021.

A. Lupu, B. Cui, H. Hu, and J. Foerster. Trajectory diversity for zero-shot coordination. In *International Conference on Machine Learning (ICML)*, pages 7204–7213, 2021.

R. Lyons. Distance covariance in metric spaces. *The Annals of Probability*, 41(5):3284–3305, 2013.

N. Mantel. The detection of disease clustering and a generalized regression approach. *Cancer Research*, 27(2):209–220, 1967.

K. R. McKee, J. Z. Leibo, C. Beattie, and R. Everett. Quantifying the effects of environment and population diversity in multi-agent reinforcement learning. *Autonomous Agents and Multi-Agent Systems*, 36(1):21, 2022.

M. Mendel and A. Naor. Expanders with respect to Hadamard spaces and random graphs. *Duke Mathematical Journal*, 164(8):1471–1548, 2015.

C. Miglioli and J. Awan. Incomplete U-statistics of equireplicate designs: Berry–Esseen bound and efficient construction. arXiv:2510.20755, 2025.

B. Miranda, A. Lee, S. Sundar, A. Casasola, R. Schaeffer, E. Obbad, and S. Koyejo. Beyond scale: The diversity coefficient as a data quality metric for variability in natural language data. arXiv:2306.13840v4, 2025.

H. K. Nandi and P. K. Sen. On the properties of U-statistics when the observations are not independent. Part Two: Unbiased estimation of the parameters of a finite population. *Calcutta Statistical Association Bulletin*, 12(4):124–148, 1963.

M. Nei and W.-H. Li. Mathematical model for studying genetic variation in terms of restriction endonucleases. *Proceedings of the National Academy of Sciences*, 76(10):5269–5273, 1979.

A. Ospanov, J. Zhang, M. Jalali, X. Cao, A. Bogdanov, and F. Farnia. Towards a scalable reference-free evaluation of generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 120892–120927, 2024.

J. Parker-Holder, A. Pacchiano, K. M. Choromanski, and S. J. Roberts. Effective diversity in population based reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 18050–18062, 2020.

A. P. Pasarkar and A. B. Dieng. Cousins of the Vendi score: A family of similarity-based diversity metrics for science and machine learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 3808–3816, 2024.

S. R. Patel and K. Jerath. Optimizing team behavior via extremum-seeking control in multi-agent reinforcement learning. In *IEEE Conference on Decision and Control (CDC)*, 2026. To appear; arXiv:2610.18808.

D. Polykovskiy, A. Zhebrak, B. Sanchez-Lengeling, S. Golovanov, O. Tatanov, S. Belyaev, R. Kurbanov, A. Artamonov, V. Aladinskiy, M. Veselov, A. Kadurin, S. Johansson, H. Chen, S. Nikolenko, A. Aspuru-Guzik, and A. Zhavoronkov. Molecular sets (MOSES): A benchmarking platform for molecular generation models. *Frontiers in Pharmacology*, 11:565644, 2020.

A. Prorok, M. A. Hsieh, and V. Kumar. The impact of diversity on optimal control policies for heterogeneous robot swarms. *IEEE Transactions on Robotics*, 33(2):346–358, 2017.

C. R. Rao. Diversity and dissimilarity coefficients: A unified approach. *Theoretical Population Biology*, 21(1):24–43, 1982.

S. Ray. Graph-SND: Sparse aggregation for behavioral diversity in multi-agent reinforcement learning. arXiv:2605.05020, 2026.

N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Conference on Empirical Methods in Natural Language Processing and International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, 2019.

C.-E. Särndal, B. Swensson, and J. Wretman. *Model Assisted Survey Sampling*. Springer Series in Statistics. Springer, New York, 1992.

A. Schrab, I. Kim, B. Guedj, and A. Gretton. Efficient aggregated kernel tests using incomplete U-statistics. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 18793–18807, 2022.

C. Schroeder de Witt, T. Gupta, D. Makoviichuk, V. Makoviychuk, P. H. S. Torr, M. Sun, and S. Whiteson. Is independent learning all you need in the StarCraft multi-agent challenge? arXiv:2011.09533, 2020.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

N. B. Shah, S. Balakrishnan, J. Bradley, A. Parekh, K. Ramchandran, and M. J. Wainwright. Estimation from pairwise comparisons: Sharp minimax bounds with topology dependence. *Journal of Machine Learning Research*, 17(58):1–47, 2016.

D. A. Spielman and N. Srivastava. Graph sparsification by effective resistances. *SIAM Journal on Computing*, 40(6):1913–1926, 2011.

K. Stasaski and M. Hearst. Semantic diversity in dialogue with natural language inference. In *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 85–98, 2022.

G. J. Székely and M. L. Rizzo. Partial distance correlation with methods for dissimilarities. *The Annals of Statistics*, 42(6):2382–2412, 2014.

G. J. Székely, M. L. Rizzo, and N. K. Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, 35(6):2769–2794, 2007.

F. Tajima. Evolutionary relationship of DNA sequences in finite populations. *Genetics*, 105 (2):437–460, 1983.

K. A. Tessera, A. Rahman, A. Storkey, and S. V. Albrecht. HyperMARL: Adaptive hypernetworks for multi-agent RL. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025.

G. Tevet and J. Berant. Evaluating the evaluation of diversity in natural language generation. In *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 326–346, 2021.

W. S. Torgerson. Multidimensional scaling: I. Theory and method. *Psychometrika*, 17(4): 401–419, 1952.

P. Twu, Y. Mostofi, and M. Egerstedt. A measure of heterogeneity in multi-agent systems. In *American Control Conference (ACC)*, pages 3972–3977, 2014.

T. Wang, H. Dong, V. Lesser, and C. Zhang. ROMA: Multi-agent reinforcement learning with emergent roles. In *International Conference on Machine Learning (ICML)*, pages 9876–9886, 2020.

T. Wang, T. Gupta, A. Mahajan, B. Peng, S. Whiteson, and C. Zhang. RODE: Learning roles to decompose multi-agent tasks. In *International Conference on Learning Representations (ICLR)*, 2021.

K. M. Wolter. *Introduction to Variance Estimation*. Springer, New York, second edition, 2007.

D. Wood, T. Mu, A. M. Webb, H. W. J. Reeve, M. Luján, and G. Brown. A unified theory of diversity in ensemble learning. *Journal of Machine Learning Research*, 24(359):1–49, 2023.

H. Xiao, K. Rasul, and R. Vollgraf. Fashion-MNIST: A novel image dataset for benchmarking machine learning algorithms. arXiv:1708.07747, 2017.

A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, et al. Qwen2.5 technical report. arXiv:2412.15115, 2024.

S. Yang, H. Hu, X. Liu, Y. Yao, E. Shi, and S. Zhang. Structured diversity control: A dual-level framework for group-aware multi-agent coordination. arXiv:2506.18651, 2025.

C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, volume 35, pages 24611–24624, 2022.

Y. Yu, H. Jiang, and Z. Lu. Informative policy representations in multi-agent reinforcement learning via joint-action distributions. arXiv:2106.05802v1, 2021. Revised version (v2, 2022) titled "Metric Policy Representations for Opponent Modeling".

R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 586–595, 2018.

X. Zhang. U-centering as subset ANOVA: Edge regression and higher-order theory. arXiv:2608.01364, 2026.

Y. Zhang, H. Diddee, S. Holm, H. Liu, X. Liu, V. Samuel, B. Wang, and D. Ippolito. NoveltyBench: Evaluating language models for humanlike diversity. arXiv:2504.05228, 2025.

L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, volume 36, pages 46595–46623, 2023.

Y. Zhong, J. G. Kuba, X. Feng, S. Hu, J. Ji, and Y. Yang. Heterogeneous-agent reinforcement learning. *Journal of Machine Learning Research*, 25(32):1–67, 2024.

J.-Y. Zhu, R. Zhang, D. Pathak, T. Darrell, A. A. Efros, O. Wang, and E. Shechtman. Toward multimodal image-to-image translation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 465–476, 2017.

Y. Zhu, S. Lu, L. Zheng, J. Guo, W. Zhang, J. Wang, and Y. Yu. Texygen: A benchmarking platform for text generation models. In *International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 1097–1100, 2018.